

Linear Accelerator (LINAC) Access Control and Interlock System (ACIS) Validation and Verification Procedure

1.7.52.1 Rev. 18

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1.0 PURPOSE

The purpose of this procedure is to validate and verify the operation of the Access Control and Interlock System (ACIS) within specifications as outlined in "LINAC Access Control and Interlock System (ACIS) Design Manual" [1].

2.0 BACKGROUND

The CLS has adopted a two-level redundant lock-up system for the Linac enclosures. The system consists of two independent chains, either two safety-rated PLCs or a single safety-rated PLC and one based on a hard-wired 24 VDC system using relay logic.[2] Fundamentally, all lock-up areas operate in the same manner, each having its own Emergency Off Switches, Digital Interlock Door Switches (SWDI) as well as Lockup Stations (LUS). Refer to the component manual [1] for more information.

2.1 Testing Philosophy

The overriding approach to the testing methodology is a meticulous and exhaustive series of tests to ensure that the system operates as required. More detail is provided in the component manual [1].

3.0 APPLICATION REQUIREMENTS

3.1 Context

This document contains the testing procedure for the LINAC ACIS. Successful completion of this version of the LINAC ACIS V&V procedure will verify and validate ACIS of LINAC Zones 1 to 5 only.

3.2 Personnel Qualification

CLSI technical personnel not involved in the design and implementation shall conduct the validation of the system. The tests require at least two personnel (referred to hereinafter as inspector 1 and inspector 2). The validation will take place in Radiological Controlled Areas (RCA) and Industrial Controlled Areas (ICA). Inspectors must have a valid radiation training module of either General Radiological Training (GRT) [4] or Radiological Worker Training (RWT) [5] and General Industrial Training (GIT) [5] as outlined in "Canadian Light Source – Facility Access" [3] to enter these areas.

In cases where the radiation sources are being tested (i.e. Booster Ring/Storage Ring Radio Frequency), appropriate personnel from Operations, Controls and Instrumentation or Engineering and Technical Services will be identified by managers of those groups to assist the inspection team with verifying system response, as required.

3.3 Supervision and Verification

The Health, Safety and Environment (HSE) manager or designate will supervise the execution of the procedure, acting either as inspector 1 or inspector 2 or simply in a supervisory capacity.

3.4 General Precautions, Limitations and Constraints

This procedure involves an individual being present in a lockup area after it has been engaged. Further precautions will be noted in the procedure section.

3.5 Prerequisites

Verification may take place after the system has been turned over from CID to HSE. The design verification includes Safety System Ring Out [7].

3.6 Personnel Protection Equipment

Personnel who enter designated construction zones during a validation procedure must wear all Personnel Protective Equipment (PPE) (i.e. work boots, safety goggles, hard hats, hearing protection, etc) that may be required. As noted in section 3.2, inspectors are expected to be familiar with precautions for entering these areas.

3.7 Equipment, Materials and Supplies

For most testing, at least two individuals are required and for the sake of convenience, wireless radio communications is recommended. A personal dosimeter must be worn at all times.

Other equipment includes:

- A voltmeter/continuity tester
- A timer, watch or stopwatch
- Pertinent drawings referenced in section 4.2, and
- Copies of the forms and checklist(s) in section 0.
- Major Fault Reset and Floor Coordinator Enable keys
- A ladder.

3.8 Termination Criteria

When a fault is detected, it is at the discretion of the HSE overseer to continue with the procedure. A fault is considered to be an incorrect response in the procedure that requires a change in wiring or software code to correct. Any fault, as defined above, generally requires that the entire validation procedure be conducted from the beginning to end after the fault has been corrected before the system is considered validated. In some cases, an exception may be permissible. Any exceptions require a memorandum identifying the fault, the proposed repair, the risk created by the repair, and the recommended validation testing to confirm the risk is mitigated. The memorandum must be reviewed, approved, and signed by the HSE and Engineering Managers or designates, and attached to the procedure for record. Incorrect results during any V&V due to procedural errors, labeling errors, or defective components that can be changed without changing the system configuration (such as light bulbs) will not require a complete revalidation of the system.

4.0 REFERENCE MATERIAL

4.1 Glossary

The following terms appear in the verification procedures:

TBN-n (Description) is a terminal block. The point is tested either by measuring voltage from the point to ground or by measuring continuity between two points, indicated by a “V” or “C” in square brackets, respectively. In the latter case, TBN-n will indicate two points separated by a “/”; TB5-13/14 or TB1-4/TB9-2, for example. Either or both points could also be a relay terminal (see Knn, this section).

Information in round brackets is a brief, optional description of the device function.

Knn (Description) is a relay, part of the ACIS (refer to wiring diagram [4]). Verification of device status at the panel is accomplished visually by observing an indicator light on the relay in question. An illuminated light denotes an energized relay represented as a 1 or “on”. An inactive light denotes a de-energized relay shown as a 0 or “off”.

In some instances it is necessary to specify a terminal on the relay to test. In such instances the terminal will be noted similar to a terminal block. For example, K3-A1+ means to test terminal “A1+” on relay 3.

Information in round brackets is a brief, optional description of the device function.

TN-n (Description) is a timer module, part of the ACIS. Verification of device status at the panel is accomplished visually by observing an indicator light on the device in question. An illuminated light denotes an energized timer, represented as a 1 or “on”. An inactive light denotes a de-energized device shown as a 0 or “off”.

In some instances it is necessary to specify a terminal on the timer to test. In such instances the terminal will be noted similar to a terminal block. For example, T2-B1 means to test terminal “B1” of timer 2.

Information in round brackets is a brief, optional description of the device function.

SWx (Description) is a test switch. These are typically normally-closed momentary pushbuttons that are used to simulate key events. In rare circumstances a normally-open contact is utilized. The information in round brackets is a brief, optional description of the device.

ALL CLEAR refers to the timed interval immediately after all Zones 1-4 are secure and the Zones 1-4 Kirk keys are inserted into the Control Room Kirk Key Bank and turned or immediately after Zone 5 is secure and the Zone 5 Kirk Key is inserted into the Control Room Kirk Key Bank and turned. During this interval the horns will sound in Zones 1-4 or Zone 5 respectively.

The following three labels refer to system components. In the context of this document it refers to whether or not the component can be operated or has been interlocked.

LINAC is the linear accelerator.

SR RF refers to the storage ring RF system

BR RF refers to the booster ring RF system.

DCN refers to a Disconnect.

4.2 Supporting Documents

1. "LINAC Access Control and Interlock System (ACIS) Design Manual", 1.9.52.1
2. "CLSI Safety Report", 11.18.40.2
3. "Canadian Light Source Facility Access", 8.7.1.1
4. "General Radiological Training (GRT)", 11.14.53.2
5. "Radiological Worker Training (RWT)", 11.14.53.1
6. "General Industrial Training", 11.14.52.2
7. "Safety System Ring Out Procedure", 7.7.52.1
8. "Linac – LTB1 Lockup Procedure", 2.7.37.2
9. "Linac, LTB1, BR1 &SR1 Zone 1 ACIS Layout", RAD\0168300
10. "Linac, LTB1, BR1 &SR1 Zone 2 ACIS Layout", RAD\0168301
11. "Linac, LTB1, BR1 &SR1 Zone 3 ACIS Layout", RAD\0168302
12. "Linac, LTB1, BR1 &SR1 Zone 4 ACIS Layout", RAD\0168303
13. "Linac, LTB1, BR1 &SR1 Zone 5 ACIS Layout", RAD\0168304
14. "ACIS Control Room P1021.3-10 Wiring Schematic",
/ACCL/EE/PROT/LOCK/0055537
15. "Linac ACIS Zone 1-5 Power Distribution Wiring Diagram"
/ACCL/EE/PROT/LOCK/0168315
16. "Linac ACIS Control Room PLC1021.4-03 Wiring Diagram, Sheet 1 of 2"
/ACCL/EE/PROT/LOCK/0168310
17. "Linac ACIS Control Room PLC1021.4-03 Wiring Diagram, Sheet 2 of 2"
/ACCL/EE/PROT/LOCK/0168311
18. "Linac ACIS Zone 1 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 1 of 3"
/ACCL/EE/PROT/LOCK/0168320
19. "Linac ACIS Zone 1 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 2 of 3"
/ACCL/EE/PROT/LOCK/0168321
20. "Linac ACIS Zone 1 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 3 of 3"
/ACCL/EE/PROT/LOCK/0168322
21. "Linac ACIS Zone 2 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 1 of 5"
/ACCL/EE/PROT/LOCK/0168325
22. "Linac ACIS Zone 2 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 2 of 5"
/ACCL/EE/PROT/LOCK/0168326
23. "Linac ACIS Zone 2 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 3 of 5"
/ACCL/EE/PROT/LOCK/0168327
24. "Linac ACIS Zone 2 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 4 of 5"
/ACCL/EE/PROT/LOCK/0168328
25. "Linac ACIS Zone 2 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 5 of 5"
/ACCL/EE/PROT/LOCK/0168329
26. "Linac ACIS Zone 3 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 1 of 3"
/ACCL/EE/PROT/LOCK/0168330

27. "Linac ACIS Zone 3 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 2 of 3" /ACCL/EE/PROT/LOCK/0168331
28. "Linac ACIS Zone 3 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 3 of 3" /ACCL/EE/PROT/LOCK/0168332
29. "Linac ACIS Zone 4 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 1 of 3" /ACCL/EE/PROT/LOCK/0168335
30. "Linac ACIS Zone 4 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 2 of 3" /ACCL/EE/PROT/LOCK/0168336
31. "Linac ACIS Zone 4 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 3 of 3" /ACCL/EE/PROT/LOCK/0168337
32. "Linac ACIS Zone 5 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 1 of 3" /ACCL/EE/PROT/LOCK/0168340
33. "Linac ACIS Zone 5 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 2 of 3" /ACCL/EE/PROT/LOCK/0168341
34. "Linac ACIS Zone 5 (P0102.5-01 & P0102.5-02) Wiring Diagram, Sheet 3 of 3" /ACCL/EE/PROT/LOCK/0168342
35. "Control Room Linac ACIS Indicator Panel (P1021.3-18) Lamacoid Layout". /ACCL/EE/PROT/LOCK/0168360
36. "Booster Lockup Procedure", 3.7.37.1
37. "Storage Ring Lockup Procedure", 5.7.52.1
38. "Booster/Storage Ring Top of Shielding Equipment Layout", /BLDG/EE/0050706
39. "Non-Conformance Reporting and Tracking Procedure", 0.7.91.2

5.0 PROCEDURE

The following procedures make use of the LINAC Lockup Procedure [8], the general layout drawings [9] to [13], BR and SR Lockup Procedures [36] [37] and the forms and checklist(s) in section 4.2. Rack locations are shown on drawing 0050706 [38].

For each non-conformance, an incident is entered into the automated problem tracking system [39] detailing the nature, cause and resolution of the non-conformance. The incident number issued by the system is listed in the summary report of section 6.1.

Notify Accelerator Operations and Development Department prior to removing any keys from control room.

5.1 Administrative Verification

NOTE: Do not proceed with subsequent sections until the following verification has been made.

5.1.1 Verification

- a) Inspector 1 or 2: RECORD the Version (from the revision control system) and the signature (as reported by the development environment) of the PLC software in the table below.

Software Version	PLC Signature and Date Software Uploaded

5.2 Relay Type and Contact Verification

Prerequisite

- a) Inspector 1 or 2: In R1021.1-09, TURN Gun Key #2 off and REMOVE Key.
b) Inspector 1 or 2: In R1032.2-01, TURN RF Drive Key off and REMOVE Key.

5.2.1 Panel 0102.5-01 Relay Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
b) Inspector 1 or 2: REMOVE HRN1021.3-05.
c) Inspector 2: In P0102.5-01, TURN off PS0102.5-01, PS0102.5-02, PS0102.5-03 and PS0102.5-04 and OPEN all fuse blocks (TB1 and TB2).
d) Inspector 1: Referring to Appendix A, VERIFY the status of the relay contacts for the P0102.5-01 devices listed in Table 1 by removing a relay (one at a time).
e) Inspector 1: Test the continuity as indicated. Place a check mark beside each verified contact condition (Closed or Open). "CLOSED" contacts have continuity while "OPEN" contacts do not. N/A (Not Applicable) Contacts are not to be tested.
f) Inspector 1: Replace that relay.
g) Perform steps d) to f) for each relay in Table 1
h) Inspector 1: VERIFY the color of the relays for the P0102.5-01 devices listed in Table 1 and place a check mark beside each verified relay color.

Note – Polarity can be an issue during testing. If a given test fails, try reversing the test leads. For the relays, in general, the positive (usually Red) lead should go to terminal 11.

P0102.5-01				
RELAYS	Terminals 11 & 12	Terminals 11 & 14	Relay Color	Part No.
K1	N/A	OPEN	Black	2966595
K2	N/A	OPEN	Black	2966595
K3	N/A	OPEN	Black	2966595
K4	N/A	OPEN	Black	2966595
K5	N/A	OPEN	Black	2966595
K6	N/A	OPEN	Black	2966595
K7	N/A	OPEN	Black	2966595
K8	N/A	OPEN	Black	2966595
K9	N/A	OPEN	Black	2966595
K10	N/A	OPEN	Black	2966595
K11	N/A	OPEN	Black	2966595
K12	N/A	OPEN	Black	2966595
K13	N/A	OPEN	Black	2966595
K14	N/A	OPEN	Black	2966595
K15	N/A	OPEN	Black	2966595
K16	N/A	OPEN	Black	2966595
K17	N/A	OPEN	Black	2966595
K18	N/A	OPEN	Black	2966595

Table 1: P0102.5-01 Relay Contact Verification

- i) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.2 Panel 0102.5-02 Relay Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
- b) Inspector 1: Referring to Appendix A, VERIFY the status of the relay contacts for the P0102.5-02 devices listed in Table 2 by removing a relay (one at a time)
- c) Inspector 1: Test the continuity as indicated. Place a check mark beside each verified contact condition (Closed or Open). "CLOSED" contacts have continuity while "OPEN" contacts do not. N/A (Not Applicable) Contacts are not to be tested.
- d) Inspector 1: Replace that relay.
- e) Perform steps c) to d) for each relay in Table 2
- f) Inspector 1: VERIFY the color of the relays for the P0102.5-02 devices listed in Table 2 and place a check mark beside each verified relay color.

Note – Polarity can be an issue during testing. If a given test fails, try reversing the test leads. For the relays, in general, the positive (usually Red) lead should go to terminal 11.

P0102.5-02				
RELAYS	Terminals 11 & 12	Terminals 11 & 14	Relay Color	Part No.
K1	N/A	OPEN	Black	2966595
K2	N/A	OPEN	Black	2966595
K3	N/A	OPEN	Black	2966595
K4	N/A	OPEN	Black	2966595
K5	N/A	OPEN	Black	2966595
K6	N/A	OPEN	Black	2966595
K7	N/A	OPEN	Black	2966595
K8	N/A	OPEN	Black	2966595
K9	N/A	OPEN	Black	2966595
K10	N/A	OPEN	Black	2966595
K11	N/A	OPEN	Black	2966595
K12	N/A	OPEN	Black	2966595
K13	N/A	OPEN	Black	2966595
K14	N/A	OPEN	Black	2966595
K15	N/A	OPEN	Black	2966595
K16	N/A	OPEN	Black	2966595
K17	N/A	OPEN	Black	2966595
K18	N/A	OPEN	Black	2966595

K19	N/A	OPEN	Black	2966595
K20	N/A	OPEN	Black	2966595
K21	N/A	OPEN	Black	2966595
K22	N/A	OPEN	Black	2966595
K23	N/A	OPEN	Black	2966595
K24	N/A	OPEN	Black	2966595
K25	N/A	OPEN	Black	2966595
K26	N/A	OPEN	Black	2966595
K27	N/A	OPEN	Black	2966595
K28	N/A	OPEN	Black	2966595
K29	N/A	OPEN	Black	2966595
K30	N/A	OPEN	Black	2966595
K31	N/A	OPEN	Black	2966595
K32	N/A	OPEN	Black	2966595
K33	N/A	OPEN	Black	2966595
K34	N/A	OPEN	Black	2966595
K35	N/A	OPEN	Black	2966595
K36	N/A	OPEN	Black	2966595
K37	N/A	OPEN	Black	2966595
K38	N/A	OPEN	Black	2966595
K39	N/A	OPEN	Black	2966595
K40	N/A	OPEN	Black	2966595
K41	N/A	OPEN	Black	2966595
K42	N/A	OPEN	Black	2966595
K43	N/A	OPEN	Black	2966595
K44	N/A	OPEN	Black	2966595
K45	N/A	OPEN	Black	2966595
K46	N/A	OPEN	Black	2966595
K47	N/A	OPEN	Black	2966595
K48	N/A	OPEN	Black	2966595
K49	N/A	OPEN	Black	2966595
K50	N/A	OPEN	Black	2966595
K51	N/A	OPEN	Black	2966595

K52	N/A	OPEN	Black	2966595
K53	N/A	OPEN	Black	2966595
K54	N/A	OPEN	Black	2966595
K55	N/A	OPEN	Black	2966595
K56	N/A	OPEN	Black	2966595
K57	N/A	OPEN	Black	2966595
K58	N/A	OPEN	Black	2966595
K59	N/A	OPEN	Black	2966595
K60	N/A	OPEN	Black	2966595
K61	N/A	OPEN	Black	2966595
K62	N/A	OPEN	Black	2966595
K63	N/A	OPEN	Black	2966595
K64	N/A	OPEN	Black	2966595
K65	N/A	OPEN	Black	2966595
K66	N/A	OPEN	Black	2966595
K67	N/A	OPEN	Black	2966595
K68	N/A	OPEN	Black	2966595
K69	N/A	OPEN	Black	2966595
K70	N/A	OPEN	Black	2966595
K71	N/A	OPEN	Black	2966595
K72	N/A	OPEN	Black	2966595
K73	N/A	OPEN	Black	2966595
K74	N/A	OPEN	Black	2966595
K75	N/A	OPEN	Black	2966595
K76	N/A	OPEN	Black	2966595

Table 2: P0102.5-02 Relay Contact Verification

g) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.3 Panel 0102.5-03 Relay Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
- b) Inspector 1: Referring to Appendix A, VERIFY the status of the relay contacts for the P0102.5-03 devices listed in Table 3 by removing a relay (one at a time)
- c) Inspector 1: Test the continuity as indicated. Place a check mark beside each verified contact condition (Closed or Open). "CLOSED" contacts have continuity while "OPEN" contacts do not. N/A (Not Applicable) Contacts are not to be tested.
- d) Inspector 1: Replace that relay.
- e) Perform steps c) to d) for each relay in Table 3
- f) Inspector 1: VERIFY the color of the relays for the P0102.5-03 devices listed in Table 3 and place a check mark beside each verified relay color.

Note – Polarity can be an issue during testing. If a given test fails, try reversing the test leads. For the relays, in general, the positive (usually Red) lead should go to terminal 11.

P0102.5-03				
RELAYS	Terminals 11 & 12	Terminals 11 & 14	Relay Color	Part No.
K1	N/A	OPEN	Black	2966595
K2	N/A	OPEN	Black	2966595
K3	Not Used			
K4	Not Used			
K5	N/A	OPEN	Black	2966595
K6	N/A	OPEN	Black	2966595
K7	N/A	OPEN	Black	2966595
K8	Not Used			
K9	Not Used			
K10	Not Used			
K11	N/A	OPEN	Black	2966595
K12	N/A	OPEN	Black	2966595
K13	N/A	OPEN	Black	2966595
K14	N/A	OPEN	Black	2966595
K15	Not Used			
K16	Not Used			
K17	N/A	OPEN	Black	2966595
K18	N/A	OPEN	Black	2966595
K19	N/A	OPEN	Black	2966595

K20	N/A	OPEN	Black	2966595
K21	N/A	OPEN	Black	2966595
K22	N/A	OPEN	Black	2966595
K23	N/A	OPEN	Black	2966595
K24	N/A	OPEN	Black	2966595
K25	N/A	OPEN	Black	2966595
K26	N/A	OPEN	Black	2966595
K27	N/A	OPEN	Black	2966595
K28	N/A	OPEN	Black	2966595
K29	N/A	OPEN	Black	2966595
K30	N/A	OPEN	Black	2966595
K31	N/A	OPEN	Black	2966595
K32	N/A	OPEN	Black	2966595
K33	N/A	OPEN	Black	2966595
K34	N/A	OPEN	Black	2966595
K35	N/A	OPEN	Black	2966595
K36	N/A	OPEN	Black	2966595
K37	N/A	OPEN	Black	2966595
K38	N/A	OPEN	Black	2966595
K39	N/A	OPEN	Black	2966595
K40	N/A	OPEN	Black	2966595
K41	N/A	OPEN	Black	2966595
K42	N/A	OPEN	Black	2966595
K43	N/A	OPEN	Black	2966595
K44	N/A	OPEN	Black	2966595
K45	N/A	OPEN	Black	2966595
K46	N/A	OPEN	Black	2966595
K47	N/A	OPEN	Black	2966595
K48	N/A	OPEN	Black	2966595
K49	N/A	OPEN	Black	2966595
K50	N/A	OPEN	Black	2966595
K51	N/A	OPEN	Black	2966595
K52	N/A	OPEN	Black	2966595

Table 3: P0102.5-03 Relay Contact Verification

g) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.4 Rack 1021.4-04 Relay Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
- b) Inspector 2: In R1021.4-04, TURN off PS1021.4-02 and OPEN all fuse blocks (TB1 and TB2).
- c) Inspector 1: Referring to Appendix A, VERIFY the status of the relay contacts for the R1021.4-04 devices listed in Table 4 by removing a relay (one at a time).
- d) Inspector 1: Test the continuity as indicated. Place a check mark beside each verified contact condition (Closed or Open). "CLOSED" contacts have continuity while "OPEN" contacts do not. N/A (Not Applicable) Contacts are not to be tested.
- e) Inspector 1: Replace that relay.
- f) Perform steps c) to e) for each relay in Table 4.
- g) Inspector 1: VERIFY the color of the relays for the R1021.4-04 devices listed in Table 4 and place a check mark beside each verified relay color.

Note – Polarity can be an issue during testing. If a given test fails, try reversing the test leads. For the relays, in general, the positive (usually Red) lead should go to terminal 11.

R1021.4-04				
RELAYS	Terminals 11 & 12	Terminals 11 & 14	Relay Color	Part No.
K1	SPARE			
K2	SPARE			
K3	SPARE			
K4	SPARE			
K5	SPARE			
K6	N/A	OPEN	Black	2966595
K7	N/A	OPEN	Black	2966595
K8	SPARE			
K9	N/A	OPEN	Black	2966595
K10	N/A	OPEN	Black	2966595
K11	N/A	OPEN	Black	2966595
K12	N/A	OPEN	Black	2966595
K13	N/A	OPEN	Black	2966595
K14	N/A	OPEN	Black	2966595
K15	N/A	OPEN	Black	2966595
K16	N/A	OPEN	Black	2966595

K17	N/A	OPEN	Black	2966595
K18	N/A	OPEN	Black	2966595
K19	N/A	OPEN	Black	2966595
K20	N/A	OPEN	Black	2966595
K21	N/A	OPEN	Black	2966595
K22	N/A	OPEN	Black	2966595
K23	N/A	OPEN	Black	2966595
K24	SPARE			
K25	CLOSED	OPEN	White	2961105
K26	Continuity test not required; relay is tested in 5.8			
K27	Continuity test not required; relay is tested in 5.8			
K28	Continuity test not required; relay is tested in 5.8			
K29	Continuity test not required; relay is tested in 5.8			

Table 4: R1021.4-04 Relay Contact Verification

h) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.5 Panel 1021.3-18 Device Contact Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
- b) Inspector 2: Toggle each of the devices in Table 5.
- c) Inspector 1: In R1021.4-04, VERIFY that continuity across the various points is toggled by placing a check in the TOGGLED column:

P1021.3-18 DEVICE	TERMINALS	TOGGLED
Major Fault Reset	TB3-60 &TB3-61	
Zones 1-4 Kirk Key Switches	TB3-52 &TB3-53	
Zones 1-4 Kirk Key Switches	TB3-54 &TB3-55	
Zone 5 Kirk Key Switches	TB3-56 &TB3-57	
Zone 5 Kirk Key Switches	TB3-58 &TB3-59	

Table 5: P1021.3-18 Device Contact Block Verification

- d) Inspector 2: In R1021.4-04, CLOSE all fuse blocks (TB1 and TB2) and TURN on PS1021.4-02.
- e) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.6 Zone 1 (Panel 0102.5-01) Device Contact Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed
- b) Inspector 1 or 2: Toggle each of the devices in Table 6.
- c) Inspector 1: VERIFY that continuity across the various points is toggled by placing a check in the TOGGLED column:

P0102.5-01 DEVICE	TERMINALS	TOGGLED
ZLL Bypass Keyswitch	TB8-119 & TB7-23	
HRN Bypass Keyswitch	TB8-120 & TB7-24	
Zone 1 P0102.5-01 DEVICE	TERMINALS	TOGGLED
SWDI00S2-01(DOOR D2)	TB7-1 & TB7-4	
SWDI00S2-02(DOOR D2)	TB8-63 & TB8-64	
SWDI0001-01 (GATE G5)	TB7-1 & TB7-6	
SWDI0001-02 (GATE G5)	TB8-67 & TB8-68	
SWDI0002-01 (GATE G3)	TB7-1 & TB7-8	
SWDI0002-02 (GATE G3)	TB8-71 & TB8-72	
SWDI0002-03 (GATE G2)	TB7-1 & TB7-10	
SWDI0002-04 (GATE G2)	TB8-75 & TB8-76	
SWDI10S2-01(DOOR D4)	TB7-1 & TB7-12	
SWDI10S2-02(DOOR D4)	TB8-79 & TB8-80	
SWDI10S2-03(DOOR D3)	TB7-1 & TB7-14	
SWDI10S2-04(DOOR D3)	TB8-83 & TB8-84	
LUS0001-01	TB7-1 & TB7-16	
LUS0002-01	TB7-1 & TB7-17	
LUS0010-01	TB7-1 & TB7-18	
LUS0011-01	TB7-1 & TB7-19	
LUS0010-02	TB7-1 & TB7-20	
EOS0002-01	TB8-1 & TB8-2	
EOS0002-01	TB8-3 & TB8-4	
EOS0002-02	TB8-7 & TB8-8	
EOS0002-02	TB8-9 & TB8-10	
EOS0002-03	TB8-13 & TB8-14	
EOS0002-03	TB8-15 & TB8-16	
EOS0001-01	TB8-19 & TB8-20	

EOS0001-01	TB8-21 & TB8-22	
EOS0010-02	TB8-25 & TB8-26	
EOS0010-02	TB8-27 & TB8-28	
EOS0001-02	TB8-31 & TB8-32	
EOS0001-02	TB8-33 & TB8-34	
EOS0001-03	TB8-37 & TB8-38	
EOS0001-03	TB8-39 & TB8-40	
EOS0001-04	TB8-43 & TB8-44	
EOS0001-04	TB8-45 & TB8-46	
EOS0010-01	TB8-49 & TB8-50	
EOS0010-01	TB8-51 & TB8-52	
EOS0010-03	TB8-55 & TB8-56	
EOS0010-03	TB8-57 & TB8-58	

Table 6: P0102.5-01 Zone 1 Device Contact Block Verification

d) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.7 Zone 2 (Panel 0102.5-02) Device Contact Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
- b) Inspector 1 or 2: Toggle each of the devices in Table 7
- c) Inspector 1: VERIFY that continuity across the various points is toggled by placing a check in the TOGGLED column:

Zone 2 P0102.5-02 DEVICE	TERMINALS	TOGGLED
SWDI0009-01 (DOOR D1)	TB1-1 & TB1-4	
SWDI0009-02 (DOOR D1)	TB5-45 & TB5-46	
SWDI0009-03 (GATE G4)	TB1-1 & TB1-6	
SWDI0009-04 (GATE G4)	TB5-45 & TB5-50	
SWDI10C1-01 (GATE G12)	TB1-1 & TB1-8	
SWDI10C1-02 (GATE G12)	TB5-45 & TB5-54	
SWDI00C1-01(DOOR D5)	TB1-1 & TB1-10	
SWDI00C1-02(DOOR D5)	TB5-45 & TB5-58	
SWDI00C1-03(DOOR D5)	TB1-1 & TB1-12	
SWDI00C1-04(DOOR D5)	TB5-45 & TB5-62	
SWDI0009-07 (GATE G8)	TB1-1 & TB1-14	
SWDI0009-08 (GATE G8)	TB5-45 & TB5-66	
SWDI0006-01 (DOOR D8)	TB1-1 & TB1-16	
SWDI0006-02 (DOOR D8)	TB5-45 & TB5-70	
SWDI0006-03 (GATE G10)	TB1-1 & TB1-18	
SWDI0006-04 (GATE G10)	TB5-45 & TB5-74	
SWDI0006-05 (GATE G10)	TB1-1 & TB1-20	
SWDI0006-06 (GATE G10)	TB5-45 & TB5-78	
SWDI0006-07 (DOOR D8)	TB1-1 & TB1-22	
SWDI0006-08 (DOOR D8)	TB5-45 & TB5-82	
SWDI0008-01 (DOOR G6)	TB1-1 & TB1-37	
SWDI0008-02 (DOOR G6)	TB5-45 & TB7-10	
SWDI00S6-01 (GATE G1)	TB1-62 & TB1-65	
SWDI00S6-02 (GATE G1)	TB6-27 & TB6-28	
SWDI0003-01 (DOOR RD1)	TB1-62 & TB1-67	
SWDI0003-02 (DOOR RD1)	TB6-27 & TB6-32	
SWDI0003-03 (GATE G7)	TB1-62 & TB1-69	

SWDI0003-04 (GATE G7)	TB6-27 & TB6-36	
SWDI01S6-01 (DOOR D9)	TB1-62 & TB1-71	
SWDI01S6-02 (DOOR D9)	TB6-27 & TB6-40	
KKSW0006-01	TB5-45 & TB7-6	
KKSW0006-01	TB1-1 & TB1-35	
LUS0003-01	TB1-62 & TB1-73	
LUS0003-02	TB1-62 & TB1-74	
LUS0003-03	TB1-62 & TB1-75	
LUS0003-04	TB1-62 & TB1-76	
LUS0003-05	TB1-62 & TB1-77	
LUS0009-01	TB1-26 & TB1-27	
LUS0009-02	TB1-26 & TB1-28	
LUS0009-03	TB1-26 & TB1-29	
LUS0006-01	TB1-26 & TB1-30	
LUS0006-02	TB1-26 & TB1-31	
LUS0007-01	TB1-26 & TB1-32	
LUS0008-01	TB1-26 & TB1-36	
EOS0009-01	TB5-1 & TB5-2	
EOS0009-01	TB5-3 & TB5-4	
EOS0009-02	TB5-7 & TB5-8	
EOS0009-02	TB5-9 & TB5-10	
EOS0009-03	TB5-13 & TB5-14	
EOS0009-03	TB5-15 & TB5-16	
EOS0009-04	TB5-19 & TB5-20	
EOS0009-04	TB5-21 & TB5-22	
EOS0009-05	TB5-25 & TB5-26	
EOS0009-05	TB5-27 & TB5-28	
EOS0006-01	TB5-31 & TB5-32	
EOS0006-01	TB5-33 & TB5-34	
EOS0006-02	TB5-37 & TB5-38	
EOS0006-02	TB5-39 & TB5-40	
EOS0003-01	TB6-1 & TB6-2	
EOS0003-01	TB6-3 & TB6-4	

EOS0003-02	TB6-7 & TB6-8	
EOS0003-02	TB6-9 & TB6-10	
EOS0003-03	TB6-13 & TB6-14	
EOS0003-03	TB6-15 & TB6-16	
EOS0003-04	TB6-19 & TB6-20	
EOS0003-04	TB6-21 & TB6-22	

Table 7: P0102.5-02 Zone 2 Device Contact Block Verification

d) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.8 Zone 3 (Panel 0102.5-03) Device Contact Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed.
- b) Inspector 1 or 2: Toggle each of the devices in Table 8.
- c) Inspector 1: VERIFY that continuity across the various points is toggled by placing a check in the TOGGLED column:

P0102.5-03 DEVICE	TERMINALS	TOGGLED
SWDI0012-03 (GATE G9)	TB1-1 & TB1-6	
SWDI0012-04 (GATE G9)	TB5-115 & TB5-120	
SWDI0013-03 (GATE G13)	TB1-1 & TB1-10	
SWDI0013-04 (GATE G13)	TB5-115 & TB5-128	
SWDI0014-01 (DOOR D7)	TB1-1 & TB1-12	
SWDI0014-02 (DOOR D7)	TB5-115 & TB6-4	
LUS0014-01	TB1-1 & TB1-16	
LUS0012-01	TB1-1 & TB1-17	
EOS0012-01	TB5-71 & TB5-72	
EOS0012-01	TB5-73 & TB5-74	
EOS0012-02	TB5-95 & TB5-96	
EOS0012-02	TB5-97 & TB5-98	
EOS0014-01	TB5-107 & TB5-108	
EOS0014-01	TB5-109 & TB5-110	

Table 8: P0102.5-03 Zone 3 Device Contact Block Verification

- d) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.9 Zone 4 (Panel 0102.5-03) Device Contact Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed
- b) Inspector 1 or 2: Toggle each of the devices in Table 9.
- c) Inspector 1: VERIFY that continuity across the various points is toggled by placing a check in the TOGGLED column:

P0102.5-03 DEVICE	TERMINALS	TOGGLED
SWDI0004-01 (DOOR D6)	TB1-37 & TB1-40	
SWDI0004-02 (DOOR D6)	TB6-97 & TB6-98	
SWDI0004-03 (GATE G11)	TB1-37 & TB1-42	
SWDI0004-04 (GATE G11)	TB6-97 & TB6-102	
SWDI0004.1-03 (GATE G6)	TB1-37 & TB1-44	
SWDI0004.1-04 (GATE G6)	TB6-97 & TB6-106	
SWDI0102.5-01 (DOOR D10)	TB1-37 & TB1-46	
SWDI0102.5-02 (DOOR D10)	TB6-97 & TB6-110	
LUS0004-01	TB1-37 & TB1-48	
LUS0004-02	TB1-37 & TB1-49	
LUS0012-02	TB1-37 & TB1-50	
LUS0004.1-01	TB1-37 & TB1-51	
LUS0004.1-02	TB1-37 & TB1-52	
LUS0102.5-01	TB1-37 & TB1-53	
EOS0004-01	TB6-53 & TB6-54	
EOS0004-01	TB6-55 & TB6-56	
EOS0004-02	TB6-59 & TB6-60	
EOS0004-02	TB6-61 & TB6-62	
EOS0004-03	TB6-65 & TB6-66	
EOS0004-03	TB6-67 & TB6-68	
EOS0004-04	TB6-71 & TB6-72	
EOS0004-04	TB6-73 & TB6-74	
EOS0004.1-01	TB6-77 & TB6-78	
EOS0004.1-01	TB6-79 & TB6-80	
EOS0004.1-02	TB6-83 & TB6-84	
EOS0004.1-02	TB6-85 & TB6-86	
EOS0102.5-01	TB6-89 & TB6-90	
EOS0102.5-01	TB6-91 & TB6-92	

Table 9: P0102.5-03 Zone 4 Device Contact Block Verification

d) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.2.10 Zone 5 (Panel 0102.5-03) Device Contact Verification

- a) Inspector 1 or 2: Ensure 5.2 Prerequisite has been completed
- b) Inspector 1 or 2: Toggle each of the devices in Table 10.
- c) Inspector 1: VERIFY that continuity across the various points is toggled by placing a check in the TOGGLED column:

P0102.5-03 DEVICE	TERMINALS	TOGGLED
SWDI0108-01 (GATE G18)	TB1-73 & TB1-76	
SWDI0108-02 (GATE G18)	TB7-93 & TB7-94	
SWDI0004.1-01 (GATE G16)	TB1-73 & TB1-78	
SWDI0004.1-02 (GATE G16)	TB7-93 & TB7-98	
SWDI1300-01 (GATE G21)	TB1-73 & TB1-80	
SWDI1300-02 (GATE G21)	TB7-93 & TB7-102	
SWDI1304-03 (GATE G19)	TB1-73 & TB1-82	
SWDI1304-04 (GATE G19)	TB7-93 & TB7-106	
SWDI1304-05 (GATE G20)	TB1-73 & TB1-84	
SWDI1304-06 (GATE G20)	TB7-93 & TB7-110	
LUS1300-01	TB1-73 & TB1-88	
LUS1300-02	TB1-73 & TB1-89	
LUS1300-03	TB1-73 & TB1-90	
LUS1300-04	TB1-73 & TB1-91	
LUS1304-05	TB1-73 & TB1-92	
LUS0108-01	TB1-73 & TB1-93	
EOS0108-01	TB7-25 & TB7-26	
EOS0108-01	TB7-27 & TB7-28	
EOS0108-02	TB7-31 & TB7-32	
EOS0108-02	TB7-33 & TB7-34	
EOS0108-03	TB7-37 & TB7-38	
EOS0108-03	TB7-39 & TB7-40	
EOS0108-04	TB7-43 & TB7-44	
EOS0108-04	TB7-45 & TB7-46	
EOS1300-01	TB7-49 & TB7-50	
EOS1300-01	TB7-51 & TB7-52	
EOS1300-02	TB7-55 & TB7-56	

EOS1300-02	TB7-57 & TB7-58	
EOS1300-03	TB7-61 & TB7-62	
EOS1300-03	TB7-63 & TB7-64	
EOS1300-04	TB7-67 & TB7-68	
EOS1300-04	TB7-69 & TB7-70	
EOS1304-07	TB7-73 & TB7-74	
EOS1304-07	TB7-75 & TB7-76	
EOS1304-08	TB7-79 & TB7-80	
EOS1304-08	TB7-81 & TB7-82	
EOS1301-01	TB7-85 & TB7-86	
EOS1301-01	TB7-87 & TB7-88	

Table 10: P0102.5-03 Zone 5 Device Contact Block Verification

- d) Inspector 2: In P0102.5-01, CLOSE all fuse blocks (TB-1 and TB-2). TURN on PS0102.5-01, PS0102.5-02, PS0102.5-03 and PS0102.5-04.
- e) At Rack R1021.4-04 Reset PLC1021.4-03 (refer to Appendix B) by turning the PLC from Run to Stop and then back to Run.
- f) Inspector 1 or 2: REPLACE HRN1021.3-05.
- g) Inspector 1 or 2: In R1021.1-09, INSERT and TURN Gun Key #2 on.
- h) Inspector 1 or 2: In R1032.2-01, INSERT and TURN RF Drive Key on.
- i) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.3 Emergency Off Verification

Prerequisite – Before performing sections 5.3.1 to 5.3.5 the following must be completed.

- a. Inspector 1 or 2: In R1021.1-09, TURN Gun Key #2 off and REMOVE Key.

- b. Inspector 1 or 2: In R1032.2-01, TURN RF Drive Key off and REMOVE Key.

- c. Inspector 1 or 2: PLACE cameras in Room 1021 for monitoring the LINAC ACIS Display Panel in P1021.3-18.

5.3.1 Zone 1 Emergency Off Verification

- a) Inspector 1 or 2: Ensure section 5.3 prerequisite is completed.
- b) Inspector 2: RESET all EOSs in Zone 1.
- c) Inspector 1 and 2: VERIFY the status of the devices in Table 11 by placing a check in the corresponding box of the table:

Zone 1 EOS Precheck		
EOS	EOS Lamp is off	EOS is installed correctly
EOS0001-01		
EOS0001-02		
EOS0001-03		
EOS0001-04		
EOS0010-01		
EOS0010-02		
EOS0010-03		
EOS0002-01		
EOS0002-02		
EOS0002-03		

Table 11: Zone 1 EOS Precheck

- d) Inspector 1: At P0102.5-01, VERIFY:
 - a. K1 & K2 are on _____
 - b. PLC0102.5-01, Module 4, CH:0 Light is on _____
 - c. P1021.3-18 Zone 1 Emergency Off indicator is off _____
- e) Inspector 2: ACTIVATE the EOSs in Zone as indicated in Table 12 and inform Inspector 1 of the status of the EOS Lamp.
- f) Inspector 1: VERIFY the status of the devices in Table 12 by placing a check in the corresponding box of the table.

Zone 1 EOS Verification				
EOS	EOS Lamp	P0102.5-01 K1 & K2	PLC0102.5-01 Module 4 CH:0 light	P1021.3-18 Zone 1 Emergency Off lamp
EOS0001-01	On	Off	Off	On
EOS0001-02	On	Off	Off	On
EOS0001-03	On	Off	Off	On
EOS0001-04	On	Off	Off	On
EOS0010-01	On	Off	Off	On
EOS0010-02	On	Off	Off	On
EOS0010-03	On	Off	Off	On
EOS0002-01	On	Off	Off	On
EOS0002-02	On	Off	Off	On
EOS0002-03	On	Off	Off	On

Table 12: Zone 1 Emergency Off Verification

- g) Inspector 2: RESET EOS and repeat steps e) to g) for the next EOS.
- h) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.3.2 Zone 2 Emergency Off Verification

- a) Inspector 1 or 2: Ensure section 5.3 prerequisite is completed.
- b) Inspector 2: RESET all EOSs in Zone 2
- c) Inspectors 1 and 2: VERIFY the status of the devices in Table 13 by placing a check in the corresponding box of the table:

Zone 2 EOS Precheck		
EOS	EOS Lamp is off	EOS is installed correctly
EOS0009-01		
EOS0009-02		
EOS0009-03		
EOS0009-04		
EOS0009-05		
EOS0006-01		
EOS0006-02		
EOS0003-01		
EOS0003-02		
EOS0003-03		
EOS0003-04		

Table 13: Zone 2 EOS Precheck

- d) Inspector 1: At P0102.5-02, VERIFY:
 - a. K27 & K28 are on _____
 - b. K57 & K58 are on _____
 - c. PLC0102.5-01, Module 6, CH:0 Light is on _____
 - d. PLC0102.5-01, Module 8, CH:0 Light is on _____
 - e. P1021.3-18 Zone 2A & 2B Emergency Off indicators are off _____
- e) Inspector 2: ACTIVATE the EOSs in Zone as indicated in Table 14 and inform Inspector 1 of the status of the EOS Lamp.
- f) Inspector 1: VERIFY the status of the devices in Table 14 by placing a check in the corresponding box of the table.

Zone 2 EOS Verification							
EOS	EOS Lamp	P0102.5-02 K27 & K28	P0102.5-02 K57 & K58	PLC0102.5-01 Module 6 CH:0 light	PLC0102.5-01 Module 8 CH:0 light	P1021.3-18 Zone 2 Emergency Off lamps	
						2A	2B
EOS0009-01	On	Off	On	Off	On	On	Off
EOS0009-02	On	Off	On	Off	On	On	Off
EOS0009-03	On	Off	On	Off	On	On	Off
EOS0009-04	On	Off	On	Off	On	On	Off
EOS0009-05	On	Off	On	Off	On	On	Off
EOS0006-01	On	Off	On	Off	On	On	Off
EOS0006-02	On	Off	On	Off	On	On	Off
EOS0003-01	On	On	Off	On	Off	Off	On
EOS0003-02	On	On	Off	On	Off	Off	On
EOS0003-03	On	On	Off	On	Off	Off	On
EOS0003-04	On	On	Off	On	Off	Off	On

Table 14: Zone 2 Emergency Off Verification

- g) Inspector 2: RESET EOS and repeat steps e) to g) for the next EOS.
- h) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.3.3 Zone 3 Emergency Off Verification

- a) Inspector 1 or 2: Ensure section 5.3 prerequisite is completed .
- b) Inspector 2: RESET all EOSs in Zone 3.
- c) Inspectors 1 and 2: VERIFY the status of the devices in Table 15 by placing a check in the corresponding box of the table:

Zone 3 EOS Precheck		
EOS	EOS Lamp is off	EOS Lamp is installed correctly
EOS0012-01		
EOS0012-02		
EOS0014-01		

Table 15: Zone 3 EOS Precheck

- d) Inspector 1: AT P0102.5-03, VERIFY:
 - a. K1 & K2 are on _____
 - b. PLC0102.5-02, Module 4, CH:0 Light is on _____
 - c. P1021.3-18 Zone 3 Emergency Off indicator is off _____
 - d. P1021.3-18 Major Fault indicator is off _____
- e) Inspector 2: ACTIVATE the EOSs in Zone as indicated in Table 16 and inform Inspector 1 of the status of the EOS Lamp.
- f) Inspector 1: VERIFY the status of the devices in Table 16 by placing a check in the corresponding box of the table.

Zone 3 EOS Verification				
EOS	EOS Lamp	P0102.5-03 K1 & K2	PLC0102.5-02 Module 4 CH:0 light	P1021.3-18 Zone 3 Emergency Off lamp
EOS0012-01	On	Off	Off	On
EOS0012-02	On	Off	Off	On
EOS0014-01	On	Off	Off	On

Table 16: Zone 3 Emergency Off Verification

- g) Inspector 2: RESET EOS and repeat steps e) to g) for the next EOS.
- h) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.3.4 Zone 4 Emergency Off Verification

- a) Inspector 1 or 2: Ensure section 5.3 prerequisite is completed. .
- b) Inspector 2: RESET all EOSs in Zone 4.
- c) Inspectors 1 and 2: VERIFY the status of the devices in Table 17 by placing a check in the corresponding box of the table:

Zone 4 EOS Precheck		
EOS	EOS Lamp is off	EOS is installed correctly
EOS0004-01		
EOS0004-02		
EOS0004-03		
EOS0004-04		
EOS0004.1-01		
EOS0004.1-02		
EOS0102.5-01		

Table 17: Zone 4 EOS Precheck

- d) Inspector 1: P0102.5-03, VERIFY:
 - a. K21 & K22 are on _____
 - b. PLC0102.5-02, Module 6, CH:0 Light is on _____
 - c. P1021.3-18 Zone 4 Emergency Off indicator is off _____
 - d. P1021.3-18 Major Fault indicator is off _____
- e) Inspector 2: ACTIVATE the EOSs in Zone as indicated in Table 18 and inform Inspector 1 of the status of the EOS Lamp.
- f) Inspector 1: VERIFY the status of the devices in Table 18 by placing a check in the corresponding box of the table.

Zone 4 EOS Verification				
EOS	EOS Lamp	P0102.5-03 K21 & K22	PLC0102.5-02 Module 6 CH:0 light	P1021.3-18 Zone 4 Emergency Off lamp
EOS0004-01	On	Off	Off	On
EOS0004-02	On	Off	Off	On
EOS0004-03	On	Off	Off	On
EOS0004-04	On	Off	Off	On
EOS0004.1-01	On	Off	Off	On
EOS0004.1-02	On	Off	Off	On
EOS0102.5-01	On	Off	Off	On

Table 18: Zone 4 Emergency Off Verification

- g) Inspector 2: RESET EOS and repeat steps e) to g) for the next EOS.
- h) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.3.5 Zone 5 Emergency Off Verification

- a) Inspector 1 or 2: Ensure section 5.3 prerequisite is completed.
- b) Inspector 2: RESET all EOSs in Zone 5
- c) Inspectors 1 and 2: VERIFY the status of the devices in **Table 19** by placing a check in the corresponding box of the table:

Zone 5 EOS Precheck		
EOS	EOS Lamp is off	EOS Lamp is installed correctly
EOS0108-01		
EOS0108-02		
EOS0108-03		
EOS0108-04		
EOS1300-01		
EOS1300-02		
EOS1300-03		
EOS1300-04		
EOS1301-01		
EOS1304-07		
EOS1304-08		

Table 19: Zone 5 EOS Precheck

- d) Inspector 1: P0102.5-03, VERIFY:
 - a. K36 & K37 are on _____
 - b. PLC0102.5-02, Module 8, CH:0 Light is on _____
 - c. P1021.3-18 Zone 5 Emergency Off indicator is off _____
 - d. P1021.3-18 Major Fault indicator is off _____
- e) Inspector 2: ACTIVATE the EOSs in Zone as indicated in Table 20 and inform Inspector 1 of the status of the EOS Lamp.
- f) Inspector 1: VERIFY the status of the devices in Table 20 by placing a check in the corresponding box of the table.

Zone 5 EOS test				
EOS	EOS Lamp	P0102.5-03 K36 & K37	PLC0102.5-02 Module 8 CH:0 light	P1021.3-18 Zone 1 Emergency Off lamp
EOS0108-01	On	Off	Off	On
EOS0108-02	On	Off	Off	On
EOS0108-03	On	Off	Off	On
EOS0108-04	On	Off	Off	On
EOS1300-01	On	Off	Off	On
EOS1300-02	On	Off	Off	On
EOS1300-03	On	Off	Off	On
EOS1300-04	On	Off	Off	On
EOS1301-01	On	Off	Off	On
EOS1304-07	On	Off	Off	On
EOS1304-08	On	Off	Off	On

Table 20: Zone 5 Emergency Off Verification

- g) Inspector 2: RESET EOS and repeat steps e) to g) for the next EOS.
- h) Inspector 1 or 2: In R1021.1-09, INSERT and TURN Gun Key #2 on.
- i) Inspector 1 or 2: In R1032.2-01, INSERT and TURN RF Drive Key on.
- j) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.4 Door/Gate Switch Verification

Prerequisite – Before performing sections 5.4.1 to 5.4.5 the following must be completed.

- a. Inspector 1 or 2: In R1021.1-09, TURN Gun Key #2 off and REMOVE Key. _____
- b. Inspector 1 or 2: In R1032.2-01, TURN RF Drive Key off and REMOVE Key.

5.4.1 Zone 1 Door/Gate Switch Verification

- a) Inspector 1 or 2: Ensure section 5.4 prerequisite is completed.
- b) Inspector 1 or 2: Ensure all EOSs are reset.
- c) Inspector 1 or 2: Ensure all Zone 1 doors/gates are closed and latched except Door 2.
- d) Inspector 2: VERIFY:
 - a. SWDI00S2-01 and SWDI00S2-02 are installed correctly _____
- e) Inspector 2: TOGGLE SWDI00S2-01.
- f) Inspector 1: VERIFY at P0102.5-01:
 - a. PLC0102.5-01, Module 4, CH:02 Light toggles _____
- g) Inspector 2: RELEASE SWDI00S2-01 and TOGGLE SWDI00S2-02.
- h) Inspector 1: VERIFY the following at P0102.5-01:
 - a. Relays K3 & K4 toggle _____
 - b. Relays K15 & K16 toggle _____
 - c. PLC0102.5-01, Module 4, CH:03 Light toggles _____
- i) Inspector 1: CLOSE Door 2.
- j) Inspector 1: TOGGLE SW1 and VERIFY:
 - a. Relay K3 toggles _____
 - b. Relay K4 stays on _____
 - c. Relays K15 & K16 toggle _____
 - d. PLC0102.5-01, Module 4, CH:03 Light stays on _____
- k) Inspector 1: TOGGLE SW2 and VERIFY:
 - a. Relay K4 toggles _____
 - b. Relay K3 stays on _____
 - c. Relays K15 & K16 stay on _____
 - d. PLC0102.5-01, Module 4, CH:03 Light toggles _____
- l) Inspector 1: TOGGLE SW3 and VERIFY:
 - a. Relay K5 toggles _____
 - b. Relay K6 stays on _____
 - c. Relays K15 & K16 toggle _____
 - d. PLC0102.5-01, Module 4, CH:05 Light stays on _____

- m) Inspector 1: TOGGLE SW4 and VERIFY:
- a. Relay K6 toggles _____
 - b. Relay K5 stays on _____
 - c. Relays K15 & K16 stay on _____
 - d. PLC0102.5-01, Module 4, CH:05 Light toggles _____
- n) Inspector 1: TOGGLE SW5 and VERIFY:
- a. Relay K7 toggles _____
 - b. Relay K8 stays on _____
 - c. Relays K15 & K16 toggle _____
 - d. PLC0102.5-01, Module 4, CH:07 Light stays on _____
- o) Inspector 1: TOGGLE SW6 and VERIFY:
- a. Relay K8 toggles _____
 - b. Relay K7 stays on _____
 - c. Relays K15 & K16 stay on _____
 - d. PLC0102.5-01, Module 4, CH:07 Light toggles _____
- p) Inspector 1: TOGGLE SW7 and VERIFY:
- a. Relay K9 toggles _____
 - b. Relay K10 stays on _____
 - c. Relays K15 & K16 toggle _____
 - d. PLC0102.5-01, Module 4, CH:09 Light stays on _____
- q) Inspector 1: TOGGLE SW8 and VERIFY:
- a. Relay K10 toggles _____
 - b. Relay K9 stays on _____
 - c. Relays K15 & K16 stay on _____
 - d. PLC0102.5-01, Module 4, CH:09 Light toggles _____
- r) Inspector 1: TOGGLE SW9 and VERIFY:
- a. Relay K11 toggles _____
 - b. Relay K12 stays on _____
 - c. Relays K15 & K16 toggle _____
 - d. PLC0102.5-01, Module 4, CH:11 Light stays on _____
- s) Inspector 1: TOGGLE SW10 and VERIFY:
- a. Relay K12 toggles _____
 - b. Relay K11 stays on _____
 - c. Relays K15 & K16 stay on _____
 - d. PLC0102.5-01, Module 4, CH:11 Light toggles _____
- t) Inspector 1: TOGGLE SW11 and VERIFY:
- a. Relay K13 toggles _____

- b. Relay K14 stays on _____
 - c. Relays K15 & K16 toggles _____
 - d. PLC0102.5-01, Module 4, CH:13 Light stays on _____
- u) Inspector 1: TOGGLE SW12 and VERIFY:
 - a. Relay K14 toggles _____
 - b. Relay K13 stays on _____
 - c. Relays K15 & K16 stay on _____
 - d. PLC0102.5-01, Module 4, CH:13 light toggles _____
- v) Inspector 1: TOGGLE SW13 and VERIFY:
 - a. PLC0102.5-01, Module 4,, CH:02 Light toggles _____
- w) Inspector 1: TOGGLE SW14 and VERIFY:
 - a. PLC0102.5-01, Module 4, CH:04 Light toggles _____
- x) Inspector 1: TOGGLE SW15 and VERIFY:
 - a. PLC0102.5-01, Module 4, CH:06 Light toggles _____
- y) Inspector 1: TOGGLE SW16 and VERIFY:
 - a. PLC0102.5-01, Module 4,, CH:08 Light toggles _____
- z) Inspector 1: TOGGLE SW17 and VERIFY:
 - a. PLC0102.5-01, Module 4, CH:10 Light toggles _____
- aa) Inspector 1: TOGGLE SW18 and VERIFY:
 - a. PLC0102.5-01, Module 4, CH:12 Light toggles _____
- bb) Inspector 1: TOGGLE P0102.5-02, SW18 and VERIFY:
 - a. Relays K15 & K16 toggle _____
- cc) Inspector 1: TOGGLE P0102.5-02, SW21 and VERIFY:
 - a. Relays K15 & K16 toggle _____
- dd) Inspector 1: TOGGLE P0102.5-02, SW53 and VERIFY:
 - a. Relays K15 & K16 toggle _____
- ee) Inspector 2: OPEN Gate 5:
- ff) Inspector 2: VERIFY:
 - a. SWDI0001-01 and SWDI0001-02 are installed correctly _____
- gg) Inspector 2: TOGGLE SWDI0001-01.
- hh) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 4, CH:04 Light toggles _____
- ii) Inspector 2: RELEASE SWDI0001-01 and TOGGLE SWDI0001-02.
- jj) Inspector 1: VERIFY:
 - a. Relays K5 & K6 toggle _____
 - b. Relays K15 & K16 toggle _____
 - c. PLC0102.5-01, Module 4, CH:05 light toggles _____

- kk) Inspector 2: CLOSE Gate 5
- ll) Inspector 2: OPEN Gate 3:
- mm) Inspector 2: VERIFY:
- a. SWDI0002-01 and SWDI0002-02 are installed correctly _____
- nn) Inspector 2: TOGGLE SWDI0002-01.
- oo) Inspector 1: VERIFY the following at P0102.5-01:
- a. PLC0102.5-01, Module 4, CH:06 Light toggles _____
- pp) Inspector 2: RELEASE SWDI0002-01 and TOGGLE SWDI0002-02.
- qq) Inspector 1: VERIFY:
- a. Relays K7 & K8 toggle _____
- b. Relays K15 & K16 toggle _____
- c. PLC0102.5-01, Module 4, CH:07 light toggles _____
- rr) Inspector 2: CLOSE Gate 3.
- ss) Inspector 2: OPEN Gate 2:
- tt) Inspector 2: VERIFY:
- a. SWDI0002-03 and SWDI0002-04 are installed correctly _____
- uu) Inspector 2: TOGGLE SWDI0002-03.
- vv) Inspector 1: VERIFY the following at P0102.5-01:
- a. PLC0102.5-01, Module 4, CH:08 Light toggles _____
- ww) Inspector 2: RELEASE SWDI0002-03 and TOGGLE SWDI0002-04.
- xx) Inspector 1: VERIFY:
- a. Relays K9 & K10 toggle _____
- b. Relays K15 & K16 toggle _____
- c. PLC0102.5-01, Module 4, CH:09 light toggles _____
- yy) Inspector 2: CLOSE Gate 2.
- zz) Inspector 2: OPEN Door 4:
- aaa) Inspector 2: VERIFY:
- a. SWDI10S2-01 and SWDI10S2-02 are installed correctly _____
- bbb) Inspector 2: TOGGLE SWDI10S2-01.
- ccc) Inspector 1: VERIFY the following at P0102.5-01:
- a. PLC0102.5-01, Module 4, CH:10 Light toggles _____
- ddd) Inspector 2: RELEASE SWDI10S2-01 and TOGGLE SWDI10S2-02.
- eee) Inspector 1: VERIFY:
- a. Relays K11 & K12 toggle _____
- b. Relays K15 & K16 toggle _____
- c. PLC0102.5-01, Module 4, CH:11 light toggles _____
- fff) Inspector 2: CLOSE Door 4.

ggg) Inspector 2: OPEN Door 3:

hhh) Inspector 2: VERIFY:

a. SWDI10S2-03 and SWDI10S2-04 are installed correctly _____

iii) Inspector 2: TOGGLE SWDI10S2-03.

jjj) Inspector 1: VERIFY the following at P0102.5-01:

a. PLC0102.5-01, Module 4, CH:12 Light toggles _____

kkk) Inspector 2: RELEASE SWDI10S2-03 and TOGGLE SWDI10S2-04.

lll) Inspector 1: VERIFY:

a. Relays K13 & K14 toggle _____

b. Relays K15 & K16 toggle _____

c. PLC0102.5-01, Module 4, CH:13 light toggles _____

mmm) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.4.2 Zone 2 Door/Gate Switch Verification

- a) Inspector 1 or 2: Ensure section 5.4 prerequisite is completed.
- b) Inspector 1 or 2: Ensure all EOSs are reset.
- c) Inspector 1 or 2: Ensure Kirk Key KKS0006-01 is enabled.
- d) Inspector 1 or 2: Ensure all Zone 2 doors/gates are closed and latched except Door D1.
- e) Inspector 2: VERIFY:
 - a. SWDI0009-01 and SWDI0009-02 are installed correctly _____
- f) Inspector 2: TOGGLE SWDI0009-01.
- g) Inspector 1: At P0102.5-01 VERIFY :
 - a. PLC0102.5-01, Module 6, CH:02 Light toggles _____
 - b. PLC0102.5-01, Module 6, CH:03 Light is off _____
- h) Inspector 2: RELEASE SWDI0009-01 and TOGGLE SWDI0009-02.
- i) Inspector 1: VERIFY the following at P0102.5-02:
 - a. Relays K29 – K31 toggle _____
 - b. Relays K53 & K54 toggle _____
 - c. PLC0102.5-01, Module 6, CH:02 Light stays off _____
 - d. PLC0102.5-01, Module 6, CH:03 Light toggles _____
- j) Inspector 1: CLOSE Door D1.
- k) Inspector 1: TOGGLE SW19 and VERIFY:
 - a. Relay K30 toggles _____
 - b. Relay K31 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:03 Light stays on _____
- l) Inspector 1: TOGGLE SW20 and VERIFY:
 - a. Relay K31 toggles _____
 - b. Relay K30 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:03 Light toggles _____
- m) Inspector 1: TOGGLE SW22 and VERIFY:
 - a. Relay K33 toggles _____
 - b. Relay K34 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:05 Light stays on _____
- n) Inspector 1: TOGGLE SW23 and VERIFY:
 - a. Relay K34 toggles _____
 - b. Relay K33 stays on _____

- c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:05 Light toggles _____
- o) Inspector 1: TOGGLE SW24 and VERIFY:
 - a. Relay K35 toggles _____
 - b. Relay K36 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:07 Light stays on _____
- p) Inspector 1: TOGGLE SW25 and VERIFY:
 - a. Relay K36 toggles _____
 - b. Relay K35 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:07 Light toggles _____
- q) Inspector 1: TOGGLE SW26 and VERIFY:
 - a. Relay K37 toggles _____
 - b. Relay K38 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:09 Light stays on _____
- r) Inspector 1: TOGGLE SW27 and VERIFY:
 - a. Relay K38 toggles _____
 - b. Relay K37 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:09 Light toggles _____
- s) Inspector 1: TOGGLE SW28 and VERIFY:
 - a. Relay K39 toggles _____
 - b. Relay K40 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:11 Light stays on _____
- t) Inspector 1: TOGGLE SW29 and VERIFY:
 - a. Relay K40 toggles _____
 - b. Relay K39 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:11 Light toggles _____
- u) Inspector 1: TOGGLE SW30 and VERIFY:
 - a. Relay K41 toggles _____
 - b. Relay K42 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:13 Light stays on _____

- v) Inspector 1: TOGGLE SW31 and VERIFY:
- a. Relay K42 toggles _____
 - b. Relay K41 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:13 Light toggles _____
- w) Inspector 1: TOGGLE SW32 and VERIFY:
- a. Relay K43 toggles _____
 - b. Relay K44 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:15 Light stays on _____
- x) Inspector 1: TOGGLE SW33 and VERIFY:
- a. Relay K44 toggles _____
 - b. Relay K43 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:15 light toggles _____
- y) Inspector 1: TOGGLE SW35 and VERIFY:
- a. Relay K46 toggles _____
 - b. Relay K47 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:17 Light stays on _____
- z) Inspector 1: TOGGLE SW36 and VERIFY:
- a. Relay K47 toggles _____
 - b. Relay K46 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:17 light toggles _____
- aa) Inspector 1: TOGGLE SW38 and VERIFY:
- a. Relay K49 toggles _____
 - b. Relay K50 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 6, CH:19 Light stays on _____
- bb) Inspector 1: TOGGLE SW39 and VERIFY:
- a. Relay K50 toggles _____
 - b. Relay K49 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:19 light toggles _____
- cc) Inspector 1: TOGGLE SW40 and VERIFY:
- a. Relay K51 toggles _____

- b. Relay K52 stays on _____
- c. Relays K53 & K54 toggle _____
- d. PLC0102.5-01, Module 6, CH:21 Light stays on _____
- dd) Inspector 1: TOGGLE SW41 and VERIFY:
 - a. Relay K52 toggles _____
 - b. Relay K51 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 6, CH:21 light toggles _____
- ee) Inspector 1: TOGGLE SW71 and VERIFY:
 - a. Relay K75 toggles _____
 - b. Relay K76 stays on _____
 - c. Relays K53 & K54 toggle _____
 - d. PLC0102.5-01, Module 7, CH:11 Light stays on _____
- ff) Inspector 1: TOGGLE SW72 and VERIFY:
 - a. Relay K76 toggles _____
 - b. Relay K75 stays on _____
 - c. Relays K53 & K54 stay on _____
 - d. PLC0102.5-01, Module 7, CH:11 light toggles _____
- gg) Inspector 1: TOGGLE P0102.5-03 SW27 and VERIFY:
 - a. P0102.5-03 Relay K25 toggles _____
 - b. P0102.5-02 Relays K53 & K54 stay on _____
 - c. P0102.5-02 Relay K69 & K70 toggle _____
- hh) Inspector 1: TOGGLE P0102.5-02 SW54 and VERIFY:
 - a. Relay K60 toggles _____
 - b. Relay K61 stays on _____
 - c. Relays K69 & K70 toggle _____
 - d. PLC0102.5-01, Module 8, CH:03 Light stays on _____
- ii) Inspector 1: TOGGLE SW55 and VERIFY:
 - a. Relay K61 toggles _____
 - b. Relay K60 stays on _____
 - c. Relays K69 & K70 stay on _____
 - d. PLC0102.5-01, Module 8, CH:03 light toggles _____
- jj) Inspector 1: TOGGLE SW56 and VERIFY:
 - a. Relay K62 toggles _____
 - b. Relay K63 stays on _____
 - c. Relays K69 & K70 toggle _____
 - d. PLC0102.5-01, Module 8, CH:05 Light stays on _____

kk) Inspector 1: TOGGLE SW57 and VERIFY:

- a. Relay K63 toggles _____
- b. Relay K62 stays on _____
- c. Relays K69 & K70 stay on _____
- d. PLC0102.5-01, Module 8, CH:05 light toggles _____

ll) Inspector 1: TOGGLE SW58 and VERIFY:

- a. Relays K53 & K54 Toggle _____
- b. Relay K64 toggles _____
- c. Relay K69 & K70 stay on _____

mm) Inspector 1: TOGGLE SW59 and VERIFY:

- a. Relay K65 toggles _____
- b. Relay K66 stays on _____
- c. Relays K69 & K70 toggle _____
- d. PLC0102.5-01, Module 8, CH:07 Light stays on _____

nn) Inspector 1: TOGGLE SW60 and VERIFY:

- a. Relay K66 toggles _____
- b. Relay K65 stays on _____
- c. Relays K69 & K70 stay on _____
- d. PLC0102.5-01, Module 8, CH:07 light toggles _____

oo) Inspector 1: TOGGLE SW61 and VERIFY:

- a. Relay K67 toggles _____
- b. Relay K68 stays on _____
- c. Relays K69 & K70 toggle _____
- d. PLC0102.5-01, Module 8, CH:09 Light stays on _____

pp) Inspector 1: TOGGLE SW62 and VERIFY:

- a. Relay K68 toggles _____
- b. Relay K67 stays on _____
- c. Relays K69 & K70 stay on _____
- d. PLC0102.5-01, Module 8, CH:09 light toggles _____

qq) Inspector 1: TOGGLE SW42 and VERIFY:

- a. PLC0102.5-01, Module 6,, CH:02 Light toggles _____

rr) Inspector 1: TOGGLE SW43 and VERIFY:

- a. PLC0102.5-01, Module 6, CH:04 Light toggles _____

ss) Inspector 1: TOGGLE SW44 and VERIFY:

- a. PLC0102.5-01, Module 6, CH:06 Light toggles _____

tt) Inspector 1: TOGGLE SW45 and VERIFY:

- a. PLC0102.5-01, Module 6,, CH:08 Light toggles _____

- uu) Inspector 1: TOGGLE SW46 and VERIFY:
 - a. PLC0102.5-01, Module 6, CH:10 Light toggles _____
- vv) Inspector 1: TOGGLE SW47 and VERIFY:
 - a. PLC0102.5-01, Module 6, CH:12 Light toggles _____
- ww) Inspector 1: TOGGLE SW48 and VERIFY:
 - a. PLC0102.5-01, Module 6, CH:14 Light toggles _____
- xx) Inspector 1: TOGGLE SW49 and VERIFY:
 - a. PLC0102.5-01, Module 6, CH:16 Light toggles _____
- yy) Inspector 1: TOGGLE SW50 and VERIFY:
 - a. PLC0102.5-01, Module 6, CH:18 Light toggles _____
- zz) Inspector 1: TOGGLE SW51 and VERIFY:
 - a. PLC0102.5-01, Module 6, CH:20 Light toggles _____
- aaa) Inspector 1: TOGGLE SW73 and VERIFY:
 - a. PLC0102.5-01, Module 7, CH:10 Light toggles _____
- bbb) Inspector 1: TOGGLE SW63 and VERIFY:
 - a. PLC0102.5-01, Module 8, CH:02 Light toggles _____
- ccc) Inspector 1: TOGGLE SW64 and VERIFY:
 - a. PLC0102.5-01, Module 8, CH:04 Light toggles _____
- ddd) Inspector 1: TOGGLE SW65 and VERIFY:
 - a. PLC0102.5-01, Module 8, CH:06 Light toggles _____
- eee) Inspector 1: TOGGLE SW66 and VERIFY:
 - a. PLC0102.5-01, Module 8,, CH:08 Light toggles _____
- fff) Inspector 2: OPEN Gate G4:
- ggg) Inspector 2: TOGGLE SWDI0009-03.
- hhh) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 6, CH:04 Light toggles _____
- iii) Inspector 2: RELEASE SWDI0009-03 and TOGGLE SWDI0009-04.
- jjj) Inspector 1: VERIFY:
 - a. Relays K32 – K34 toggle _____
 - b. Relays K53 & K54 toggle _____
 - c. PLC0102.5-01, Module 6, CH:05 light toggles _____
- kkk) Inspector 2: CLOSE Gate G4
- lll) Inspector 2: OPEN Gate G12:
- mmm) Inspector 2: TOGGLE SWDI10C1-01.
- nnn) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 6, CH:06 Light toggles _____
- ooo) Inspector 2: RELEASE SWDI10C1-01 and TOGGLE SWDI10C1-02.

ppp) Inspector 1: VERIFY:

- a. Relays K35 & K36 toggle _____
- b. Relays K53 & K54 toggle _____
- c. PLC0102.5-01, Module 6, CH:07 light toggles _____

qqq) Inspector 2: CLOSE Gate G12.

rrr) Inspector 2: OPEN Door D5:

sss) Inspector 2: TOGGLE SWDI00C1-01.

ttt) Inspector 1: VERIFY the following at P0102.5-01:

- a. PLC0102.5-01, Module 6, CH:08 Light toggles _____

uuu) Inspector 2: RELEASE SWDI00C1-01 and TOGGLE SWDI00C1-02.

vvv) Inspector 1: VERIFY:

- a. Relays K37 & K38 toggle _____
- b. Relays K53 & K54 toggle _____
- c. PLC0102.5-01, Module 6, CH:09 light toggles _____

www) Inspector 2: CLOSE Door D5.

xxx) Inspector 2: OPEN Door D5:

yyy) Inspector 2: TOGGLE SWDI00C1-03.

zzz) Inspector 1: VERIFY the following at P0102.5-01:

- a. PLC0102.5-01, Module 6, CH:10 Light toggles _____

aaaa) Inspector 2: RELEASE SWDI00C1-03 and TOGGLE SWDI00C1-04.

bbbb) Inspector 1: VERIFY:

- a. Relays K39 & K40 toggle _____
- b. Relays K53 & K54 toggle _____
- c. PLC0102.5-01, Module 6, CH:11 light toggles _____

cccc) Inspector 2: CLOSE Door D5.

dddd) Inspector 2: OPEN Gate G8:

eeee) Inspector 2: TOGGLE SWDI0009-07.

fff) Inspector 1: VERIFY the following at P0102.5-01:

- a. PLC0102.5-01, Module 6, CH:12 Light toggles _____

gggg) Inspector 2: RELEASE SWDI0009-07 and TOGGLE SWDI0009-08.

hhhh) Inspector 1: VERIFY:

- a. Relays K41 & K42 toggle _____
- b. Relays K53 & K54 toggle _____
- c. PLC0102.5-01, Module 6, CH:13 light toggles _____

iiii) Inspector 2: CLOSE Gate G8.

jjjj) Inspector 2: OPEN Door D8 Right:

kkkk) Inspector 2: TOGGLE SWDI0006-01.

lll) Inspector 1: VERIFY the following at P0102.5-01:

a. PLC0102.5-01, Module 6, CH:14 Light toggles _____

mmmm) Inspector 2: RELEASE SWDI0006-01 and TOGGLE SWDI0006-02.

nnnn) Inspector 1: VERIFY:

a. Relays K43 & K44 toggle _____

b. Relays K53 & K54 toggle _____

c. PLC0102.5-01, Module 6, CH:15 light toggles _____

NOTE: Door 8 and Gate 10 each have two doors/gates. The Door/Gate that is on the Left when being pulled open is designated as the Left Door/Gate and the other door/gate is the Right Door/Gate.

oooo) Inspector 2: CLOSE Door D8 Right.

pppp) Inspector 2: OPEN Door D8 Left:

qqqq) Inspector 2: TOGGLE SWDI0006-07.

rrrr) Inspector 1: VERIFY the following at P0102.5-01:

a. PLC0102.5-01, Module 6, CH:20 Light toggles _____

ssss) Inspector 2: RELEASE SWDI0006-07 and TOGGLE SWDI0006-08.

tttt) Inspector 1: VERIFY:

a. Relays K51 & K52 toggle _____

b. Relays K53 & K54 toggle _____

c. PLC0102.5-01, Module 6, CH:21 light toggles _____

uuuu) Inspector 2: CLOSE Door D8 Left.

vvvv) Inspector 2: OPEN Gate G10 Left:

wwwv) Inspector 2: TOGGLE SWDI0006-03.

xxxx) Inspector 1: VERIFY the following at P0102.5-01:

a. PLC0102.5-01, Module 6, CH:16 Light toggles _____

yyyy) Inspector 2: RELEASE SWDI0006-03 and TOGGLE SWDI0006-04.

zzzz) Inspector 1: VERIFY:

a. Relays K45 - K47 toggle _____

b. Relays K53 & K54 toggle _____

c. PLC0102.5-01, Module 6, CH:17 light toggles _____

aaaaa) Inspector 2: CLOSE Gate G10 Left.

bbbbb) Inspector 2: OPEN Gate G10 Right.

ccccc) Inspector 2: TOGGLE SWDI0006-05.

ddddd) Inspector 1: VERIFY the following at P0102.5-01:

a. PLC0102.5-01, Module 6, CH:18 Light toggles _____

eeee) Inspector 2: RELEASE SWDI0006-05 and TOGGLE SWDI0006-06.

ffff) Inspector 1: VERIFY:

a. Relays K48 – K50 toggle _____

- b. Relays K53 & K54 toggle _____
 - c. PLC0102.5-01, Module 6, CH:19 light toggles _____
- ggggg) Inspector 2: CLOSE Gate G10 Right.
- hhhhh) Inspector 2: OPEN Gate G6.
- iiii) Inspector 2: TOGGLE SWDI0008-01.
- jjjj) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 7, CH:10 Light toggles _____
- kkkkk) Inspector 2: RELEASE SWDI0008-01 and TOGGLE SWDI0008-02.
- llll) Inspector 1: VERIFY:
 - a. Relays K75 & K76 toggle _____
 - b. Relays K53 & K54 toggle _____
 - c. PLC0102.5-01, Module 7, CH:11 light toggles _____
- mmmmm) Inspector 2: CLOSE Gate G6.
- nnnnn) Inspector 2: OPEN Gate G1:
- ooooo) Inspector 2: TOGGLE SWDI00S6-01.
- ppppp) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 8, CH:02 Light toggles _____
- qqqqq) Inspector 2: RELEASE SWDI00S6-01 and TOGGLE SWDI00S6-02.
- rrrrr) Inspector 1: VERIFY:
 - a. Relays K59 - K61 toggle _____
 - b. Relays K69 & K70 toggle _____
 - c. PLC0102.5-01, Module 8, CH:03 light toggles _____
- sssss) Inspector 2: CLOSE Gate G1.
- ttttt) Inspector 2: OPEN Door RD1:
- uuuuu) Inspector 2: TOGGLE SWDI0003-01.
- vvvvv) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 8, CH:04 Light toggles _____
- wwwww) Inspector 2: RELEASE SWDI0003-01 and TOGGLE SWDI0003-02.
- xxxxx) Inspector 1: VERIFY:
 - a. Relays K62 & K63 toggle _____
 - b. Relays K69 & K70 toggle _____
 - c. PLC0102.5-01, Module 8, CH:05 light toggles _____
- yyyyy) Inspector 2: CLOSE Door RD1.
- zzzzz) Inspector 2: OPEN Gate G7:
- aaaaa) Inspector 2: TOGGLE SWDI0003-03.
- bbbbbb) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-01, Module 8, CH:06 Light toggles _____

ccccc) Inspector 2: RELEASE SWDI0003-03 and TOGGLE SWDI0003-04.

dddddd) Inspector 1: VERIFY:

- a. Relays K64 – K66 toggle _____
- b. Relays K69 & K70 toggle _____
- c. PLC0102.5-01, Module 8, CH:07 light toggles _____

eeeeee) Inspector 2: Close Gate 7

fffff) Inspector 2: OPEN Door 9:

gggggg) Inspector 2: TOGGLE SWDI01S6-01.

hhhhh) Inspector 1: VERIFY the following at P0102.5-01:

- a. PLC0102.5-01, Module 8, CH:08 Light toggles _____

iiiiii) Inspector 2: RELEASE SWDI01S6-01 and TOGGLE SWDI01S6-02.

jjjjj) Inspector 1: VERIFY:

- a. Relays K67 – K70 toggle _____
- b. PLC0102.5-01, Module 8, CH:09 light toggles _____

kkkkk) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.4.3 Zone 3 Door/Gate Switch Verification

- a) Inspector 1 or 2: Ensure section 5.4 prerequisite is completed.
- b) Inspector 1 or 2: Ensure all EOSs are reset.
- c) Inspector 1 or 2: Ensure all Zone 3 doors/gates are closed and latched.
- d) Inspector 1: At P0102.5-03, TOGGLE SW3 and VERIFY:
 - a. Relay K5 toggles _____
 - b. Relays K6 & K7 stay on _____
 - c. Relays K17 & K18 stay on _____
 - d. PLC0102.5-02, Module 4, CH:05 Light stays on _____
- e) Inspector 1: TOGGLE SW4 and VERIFY:
 - a. Relay K6 toggles _____
 - b. Relays K5 & K7 stay on _____
 - c. Relays K17 & K18 toggle _____
 - d. PLC0102.5-02, Module 4, CH:05 Light stays on _____
- f) Inspector 1: TOGGLE SW5 and VERIFY:
 - a. Relay K7 toggles _____
 - b. Relays K5 & K6 stay on _____
 - c. Relays K17 & K18 stay on _____
 - d. PLC0102.5-02, Module 4, CH:05 Light toggles _____
- g) Inspector 1: TOGGLE SW9 and VERIFY:
 - a. Relay K11 toggles _____
 - b. Relay K12 stays on _____
 - c. Relays K17 & K18 toggle _____
 - d. PLC0102.5-02, Module 4, CH:09 Light stays on _____
- h) Inspector 1: TOGGLE SW10 and VERIFY:
 - a. Relay K12 toggles _____
 - b. Relay K11 stays on _____
 - c. Relays K17 & K18 stay on _____
 - d. PLC0102.5-02, Module 4, CH:09 Light toggles _____
- i) Inspector 1: TOGGLE SW11 and VERIFY:
 - a. Relay K13 toggles _____
 - b. Relay K14 stays on _____
 - c. Relays K17 & K18 toggle _____
 - d. PLC0102.5-02, Module 4, CH:11 Light stays on _____
- j) Inspector 1: TOGGLE SW12 and VERIFY:
 - a. Relay K14 toggles _____

- b. Relay K13 stays on _____
 - c. Relays K17 & K18 stay on _____
 - d. PLC0102.5-02, Module 4, CH:11 light toggles _____
- k) Inspector 1: TOGGLE SW16 and VERIFY:
 - a. PLC0102.5-02, Module 4, CH:04 Light toggles _____
- l) Inspector 1: TOGGLE SW18 and VERIFY:
 - a. PLC0102.5-02, Module 4, CH:08 Light toggles _____
- m) Inspector 1: TOGGLE SW19 and VERIFY:
 - a. PLC0102.5-02, Module 4, CH:10 Light toggles _____
- n) Inspector 2: OPEN Door 7:
- o) Inspector 2: TOGGLE SWDI0014-01.
- p) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 4, CH:10 Light toggles _____
- q) Inspector 2: RELEASE SWDI0014-01 and TOGGLE SWDI0014-02.
- r) Inspector 1: VERIFY:
 - a. Relays K13 & K14 toggle _____
 - b. Relays K17 & K18 toggle _____
 - c. PLC0102.5-02, Module 4, CH:11 light toggles _____
- s) Inspector 2: CLOSE Door 7:
- t) Inspector 2: OPEN Gate 9:
- u) Inspector 2: TOGGLE SWDI0012-03.
- v) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 4, CH:04 Light toggles _____
- w) Inspector 2: RELEASE SWDI0012-03 and TOGGLE SWDI0012-04.
- x) Inspector 1: VERIFY:
 - a. Relays K5 – K7 toggle _____
 - b. Relays K17 & K18 toggle _____
 - c. PLC0102.5-02, Module 4, CH:05 light toggles _____
- y) Inspector 2: CLOSE Gate 9
- z) Inspector 2: OPEN Gate 13:
- aa) Inspector 2: TOGGLE SWDI0013-03.
- bb) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 4, CH:08 Light toggles _____
- cc) Inspector 2: RELEASE SWDI0013-03 and TOGGLE SWDI0013-04.
- dd) Inspector 1: VERIFY:
 - a. Relays K11 & K12 toggle _____
 - b. Relays K17 & K18 toggle _____

- c. PLC0102.5-02, Module 4, CH:09 light toggles _____
- ee) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.4.4 Zone 4 Door/Gate Switch Verification

- a) Inspector 1 or 2: Ensure section 5.4 prerequisite is completed.
- b) Inspector 1 or 2: Ensure all EOSs are reset.
- c) Inspector 1 or 2: Ensure all Zone 4 doors/gates are closed and latched except Door 6.
- d) Inspector 2: VERIFY:
 - a. SWDI0004-01 and SWDI0004-02 are installed correctly _____
- e) Inspector 2: TOGGLE SWDI0004-01.
- f) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 6, CH:02 Light toggles _____
- g) Inspector 2: RELEASE SWDI0004-01 and TOGGLE SWDI0004-02.
- h) Inspector 1: VERIFY the following at P0102.5-03:
 - a. Relays K23 & K24 toggle _____
 - b. Relays K32 & K33 toggle _____
 - c. PLC0102.5-02, Module 6, CH:03 Light toggles _____
- i) Inspector 1: CLOSE Door 6.
- j) Inspector 1: TOGGLE SW25 and VERIFY:
 - a. Relay K23 toggles _____
 - b. Relay K24 stays on _____
 - c. Relays K32 & K33 toggle _____
 - d. PLC0102.5-02, Module 6, CH:03 Light stays on _____
- k) Inspector 1: TOGGLE SW26 and VERIFY:
 - a. Relay K24 toggles _____
 - b. Relay K23 stays on _____
 - c. Relays K32 & K33 stay on _____
 - d. PLC0102.5-02, Module 6, CH:03 Light toggles _____
- l) Inspector 1: TOGGLE SW28 and VERIFY:
 - a. Relay K26 toggles _____
 - b. Relay K25 & K27 stay on _____
 - c. Relays K32 & K33 toggle _____
 - d. PLC0102.5-02, Module 6, CH:05 Light stays on _____
- m) Inspector 1: TOGGLE SW29 and VERIFY:
 - a. Relay K27 toggles _____
 - b. Relay K25 & K26 stay on _____
 - c. Relays K32 & K33 stay on _____
 - d. PLC0102.5-02, Module 6, CH:05 Light toggles _____

- n) Inspector 1: TOGGLE SW30 and VERIFY:
 - a. Relay K28 toggles _____
 - b. Relay K29 stays on _____
 - c. Relays K32 & K33 toggle _____
 - d. PLC0102.5-02, Module 6, CH:07 Light stays on _____
- o) Inspector 1: TOGGLE SW31 and VERIFY:
 - a. Relay K29 toggles _____
 - b. Relay K28 stays on _____
 - c. Relays K32 & K33 stay on _____
 - d. PLC0102.5-02, Module 6, CH:07 Light toggles _____
- p) Inspector 1: TOGGLE SW32 and VERIFY:
 - a. Relay K30 toggles _____
 - b. Relay K31 stays on _____
 - c. Relays K32 & K33 toggle _____
 - d. PLC0102.5-02, Module 6, CH:09 Light stays on _____
- q) Inspector 1: TOGGLE SW33 and VERIFY:
 - a. Relay K31 toggles _____
 - b. Relay K30 stays on _____
 - c. Relays K32 & K33 stay on _____
 - d. PLC0102.5-02, Module 6, CH:09 Light toggles _____
- r) Inspector 1: TOGGLE SW34 and VERIFY:
 - a. PLC0102.5-02, Module 6,, CH:02 Light toggles _____
- s) Inspector 1: TOGGLE SW35 and VERIFY:
 - a. PLC0102.5-02, Module 6, CH:04 Light toggles _____
- t) Inspector 1: TOGGLE SW36 and VERIFY:
 - a. PLC0102.5-02, Module 6, CH:06 Light toggles _____
- u) Inspector 1: TOGGLE SW37 and VERIFY:
 - a. PLC0102.5-02, Module 6, CH:08 Light toggles _____
- v) Inspector 1: At P0102.5-02, TOGGLE SW34 and VERIFY:
 - a. At P0102.5-03 Relays K32 & K33 toggle _____
- w) Inspector 1: At P0102.5-02, TOGGLE SW37 and VERIFY:
 - a. At P0102.5-03 Relays K32 & K33 toggle _____
- x) Inspector 1: At P0102.5-03, TOGGLE SW41 and VERIFY:
 - a. Relays K32 & K33 toggle _____
- y) Inspector 2: OPEN Gate 11:
- z) Inspector 2: TOGGLE SWDI0004-03.
- aa) Inspector 1: VERIFY the following at P0102.5-01:

- a. PLC0102.5-02, Module 6, CH:04 Light toggles _____
- bb) Inspector 2: RELEASE SWDI0004-03 and TOGGLE SWDI0004-04.
- cc) Inspector 1: VERIFY:
- a. Relays K25 – K27 toggle _____
- b. Relays K32 & K33 toggle _____
- c. PLC0102.5-02, Module 6, CH:05 light toggles _____
- dd) Inspector 2: CLOSE Gate11
- ee) Inspector 2: OPEN Gate 6:
- ff) Inspector 2: TOGGLE SWDI0004.1-03.
- gg) Inspector 1: VERIFY the following at P0102.5-01:
- a. PLC0102.5-02, Module 6, CH:06 Light toggles _____
- hh) Inspector 2: RELEASE SWDI0004.1-03 and TOGGLE SWDI0004.1-04.
- ii) Inspector 1: VERIFY:
- a. Relays K28 & K29 toggle _____
- b. Relays K32 & K33 toggle _____
- c. PLC0102.5-02, Module 6, CH:07 light toggles _____
- jj) Inspector 2: CLOSE Gate 6
- kk) Inspector 2: OPEN Door 10:
- ll) Inspector 2: TOGGLE SWDI0102.5-01.
- mm) Inspector 1: VERIFY the following at P0102.5-01:
- a. PLC0102.5-02, Module 6, CH:08 Light toggles _____
- nn) Inspector 2: RELEASE SWDI0102.5-01 and TOGGLE SWDI0102.5-02.
- oo) Inspector 1: VERIFY:
- a. Relays K30 – K33 toggle _____
- b. PLC0102.5-02, Module 6, CH:09 light toggles _____
- pp) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.4.5 Zone 5 Door/Gate Switch Verification

- a) Inspector 1 or 2: Ensure section 5.4 prerequisite is completed.
- b) Inspector 1 or 2: Ensure all EOSs are reset.
- c) Inspector 1 or 2: Ensure all Zone 5 doors/gates are closed and latched except Gate G18.
- d) Inspector 2: VERIFY:
 - a. SWDI0108-01 and SWDI0108-02 are installed correctly _____
- e) Inspector 2: TOGGLE SWDI0108-01.
- f) Inspector 1: VERIFY at P0102.5-01:
 - a. PLC0102.5-02, Module 8, CH:02 Light toggles _____
- g) Inspector 2: RELEASE SWDI0108-01 and TOGGLE SWDI0108-02.
- h) Inspector 1: VERIFY the following at P0102.5-03:
 - a. Relays K38 & K39 toggle _____
 - b. Relays K49 & K50 toggle _____
 - c. PLC0102.5-02, Module 8, CH:03 Light toggles _____
- i) Inspector 1: CLOSE Gate G18.
- j) Inspector 1: TOGGLE SW39 and VERIFY:
 - a. Relay K38 toggles _____
 - b. Relay K39 stays on _____
 - c. Relays K49 & K50 toggle _____
 - d. PLC0102.5-02, Module 8, CH:03 Light stays on _____
- k) Inspector 1: TOGGLE SW40 and VERIFY:
 - a. Relay K39 toggles _____
 - b. Relay K38 stays on _____
 - c. Relays K49 & K50 stay on _____
 - d. PLC0102.5-02, Module 8, CH:03 Light toggles _____
- l) Inspector 1: TOGGLE SW41 and VERIFY:
 - a. Relay K40 toggles _____
 - b. Relays K41 & K42 stay on _____
 - c. Relays K49 & K50 stay on _____
 - d. PLC0102.5-02, Module 8, CH:05 Light stays on _____
- m) Inspector 1: TOGGLE SW42 and VERIFY:
 - a. Relay K41 toggles _____
 - b. Relays K40 & K42 stay on _____
 - c. Relays K49 & K50 toggle _____
 - d. PLC0102.5-02, Module 8, CH:05 Light stays on _____

- n) Inspector 1: TOGGLE SW43 and VERIFY:
- a. Relay K42 toggles _____
 - b. Relays K40 & K41 stay on _____
 - c. Relays K49 & K50 stay on _____
 - d. PLC0102.5-02, Module 8, CH:05 Light toggles _____
- o) Inspector 1: TOGGLE SW44 and VERIFY:
- a. Relay K43 toggles _____
 - b. Relay K44 stays on _____
 - c. Relays K49 & K50 toggle _____
 - d. PLC0102.5-02, Module 8, CH:07 Light stays on _____
- p) Inspector 1: TOGGLE SW45 and VERIFY:
- a. Relay K44 toggles _____
 - b. Relay K43 stays on _____
 - c. Relays K49 & K50 stay on _____
 - d. PLC0102.5-02, Module 8, CH:07 Light toggles _____
- q) Inspector 1: TOGGLE SW46 and VERIFY:
- a. Relay K45 toggles _____
 - b. Relay K46 stays on _____
 - c. Relays K49 & K50 toggle _____
 - d. PLC0102.5-02, Module 8, CH:09 Light stays on _____
- r) Inspector 1: TOGGLE SW47 and VERIFY:
- a. Relay K46 toggles _____
 - b. Relay K45 stays on _____
 - c. Relays K49 & K50 stay on _____
 - d. PLC0102.5-02, Module 8, CH:09 Light toggles _____
- s) Inspector 1: TOGGLE SW48 and VERIFY:
- a. Relay K47 toggles _____
 - b. Relay K48 stays on _____
 - c. Relays K49 & K50 toggle _____
 - d. PLC0102.5-02, Module 8, CH:11 Light stays on _____
- t) Inspector 1: TOGGLE SW49 and VERIFY:
- a. Relay K48 toggles _____
 - b. Relay K47 stays on _____
 - c. Relays K49 & K50 stay on _____
 - d. PLC0102.5-02, Module 8, CH:11 Light toggles _____
- u) Inspector 1: TOGGLE SW50 and VERIFY:
- a. PLC0102.5-02, Module 8, CH:02 Light toggles _____

- v) Inspector 1: TOGGLE SW51 and VERIFY:
 - a. PLC0102.5-02, Module 8, CH:04 Light toggles _____
- w) Inspector 1: TOGGLE SW52 and VERIFY:
 - a. PLC0102.5-02, Module 8,, CH:06 Light toggles _____
- x) Inspector 1: TOGGLE SW53 and VERIFY:
 - a. PLC0102.5-02, Module 8, CH:08 Light toggles _____
- y) Inspector 1: TOGGLE SW54 and VERIFY:
 - a. PLC0102.5-02, Module 8, CH:10 Light toggles _____
- z) Inspector 2: OPEN Gate 16:
- aa) Inspector 2: TOGGLE SWDI0004.1-01.
- bb) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 8, CH:04 Light toggles _____
- cc) Inspector 2: RELEASE SWDI0004.1-01 and TOGGLE SWDI0004.1-02.
- dd) Inspector 1: VERIFY:
 - a. Relays K40 – K42 toggle_____
 - b. Relays K49 & K50 toggle _____
 - c. PLC0102.5-02, Module 8, CH:05 light toggles _____
- ee) Inspector 2: CLOSE Gate 16.
- ff) Inspector 2: OPEN Gate 21:
- gg) Inspector 2: TOGGLE SWDI1300-01.
- hh) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 8, CH:06 Light toggles _____
- ii) Inspector 2: RELEASE SWDI1300-01 and TOGGLE SWDI1300-02.
- jj) Inspector 1: VERIFY:
 - a. Relays K43 & K44 toggle_____
 - b. Relays K49 & K50 toggle _____
 - c. PLC0102.5-02, Module 8, CH:07 light toggles _____
- kk) Inspector 2: CLOSE Gate 21.
- ll) Inspector 2: OPEN Gate 19:
- mm) Inspector 2: TOGGLE SWDI1300-03.
- nn) Inspector 1: VERIFY the following at P0102.5-01:
 - a. PLC0102.5-02, Module 8, CH:08 Light toggles _____
- oo) Inspector 2: RELEASE SWDI1300-03 and TOGGLE SWDI1300-04.
- pp) Inspector 1: VERIFY:
 - a. Relays K45 & K46 toggle_____
 - b. Relays K49 & K50 toggle _____
 - c. PLC0102.5-02, Module 8, CH:09 light toggles _____

- qq) Inspector 2: CLOSE Gate 19.
- rr) Inspector 2: OPEN Gate 20:
- ss) Inspector 2: TOGGLE SWDI1304-05.
- tt) Inspector 1: VERIFY the following at P0102.5-01:
- a. PLC0102.5-02, Module 8, CH:10 Light toggles _____
- uu) Inspector 2: RELEASE SWDI1304-05 and TOGGLE SWDI1304-06.
- vv) Inspector 1: VERIFY:
- a. Relays K47 & K48 toggle _____
- b. Relays K49 & K50 toggle _____
- c. PLC0102.5-02, Module 8, CH:11 light toggles _____
- ww) Inspector 1 or 2: In R1021.1-09, INSERT and TURN Gun Key #2 on.
- xx) Inspector 1 or 2: In R1032.2-01, INSERT and TURN RF Drive Key on.
- yy) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.5 Zone Lockup Light and Horns

Note – Refer to checklist below.

Note – Hearing protection is required for Inspector 2.

- a) Inspector 1 and 2: PLACE signs indicating ACIS testing in progress at zone 1 – 5 entrances. ENSURE that all of zones 1-5 are locked up, including turning all gate keys in the control room and allowing the respective All Clear intervals to time out.
- b) Inspector 1: At P0102.5-01, TURN the Horn Bypass and Zone Lockup Light Bypass keyswitches to the "BYPASS" position.
- c) Inspector 2: At the LINAC ACIS DISPLAY PANEL (P1021.3-18), ENSURE the "LOCKED-UP" lamps for all zones turn off. _____
- d) Inspector 1 and 2: UNLOCK and PROCEED through Zones 1 to 5, and place a checkmark beside each properly functioning device on the checklist.
- e) Inspector 1 and 2: For each zone, establish perimeter shielding. Place a checkmark in the table where lockup cannot be started in light/horn bypass mode. This can be done by activating lockup station#1 and ensuring the LUS lamp remains off.
- f) Inspector 1: At P0102.5-01, TURN the Horn Bypass and Zone Lockup Light Bypass keyswitches to the "NORMAL" position.
- g) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Zone 1					
ZLL0001-01		HRN0001-01		Cannot Lockup	
ZLL0002-01		HRN0002-01			
ZLL0010-01		HRN0010-01			
Zone 2A					
ZLL0009-01		HRN0009-01		Cannot Lockup	
ZLL0009-02		HRN0009-02			
ZLL0009-03					
ZLL10C1-01					
ZLL00C1-01					
ZLL00S1-01					
ZLL0006-01		HRN0006-01			
Zone 2B					
ZLL0003-01		HRN0003-01		Cannot Lockup	
ZLL0003-02		HRN0003-02			
Zone 3					
ZLL0012-01		HRN0012-01		Cannot Lockup	
ZLL0012-02		HRN0012-02			
ZLL0014-01					

Zone 4					
ZLL0004-01		HRN0004-01		Cannot Lockup	
ZLL0004-02		HRN0004-02			
ZLL0012-03					
ZLL0102.5-01					
ZLL0004.1-01					
Zone 5					
ZLL1300-01		HRN1300-01		Cannot Lockup	
ZLL1300-03					
ZLL1301-01					
ZLL1304-05					
ZLL0108-01		HRN0108-01			

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.6 Normal Lockup Verification

NOTE: This section refers to all of the steps between the “Normal Lockup” and “Area Compromise” banners of Tables 21 - 26.

NOTE: References to zone 2 are intended to be inclusive of both zone 2A and zone 2B.

NOTE: Before starting this section of the procedure make sure that all areas of zones 1-5 are unlocked.

NOTE: During this test, Inspector 2 will be in each zone during the All Clear interval when the horns are active and is to wear appropriate hearing protection.

- a) Inspector 1 or 2: In R1021.1-09, TURN Gun Key #2 off and REMOVE Key.
- b) Inspector 1 or 2: In R1032.2-01, TURN RF Drive Key off and REMOVE Key.
- c) Inspector 2: Perform the lockups [8] of the various zones (1-5) as shown in the Tables 21 – 26.
- d) Inspector 1 & 2: For each mirror listed in Tables 21 – 26, place a checkmark if the mirror is installed, labeled, and provides an appropriate view of the intended area.
- e) Inspector 1 & 2: As each stage is completed as indicated by the checklist, VERIFY the status of the devices as indicated in Tables 21 – 26.
- f) Inspector 1 & 2: Perform the lockup of zones 1-4 with Inspector 2 remaining inside Zone 1.
- g) Inspector 1: At P1021.3-18 Insert Zones 1-4 Kirk Keys and TURN.
- h) Inspector 2: VERIFY:
 - a. Zone 1 All Clear horns are sounding _____
- i) Inspector 1: After the all clear interval, REMOVE Zone 1 and 2A kirk keys.
- j) Inspector 2: Break lockup using the device indicated in the table.
- k) Inspector 1 & 2: VERIFY the status of the devices as indicated in Table 21.
- l) Inspector 1 & 2: Perform the lockup of zones 1-4 with Inspector 2 remaining inside Zone 2A.
- m) Inspector 1: AT P1021.3-18 Insert Zones 1-4 Kirk Keys and TURN.
- n) Inspector 2: VERIFY:
 - a. Zone 2A All Clear horns are sounding _____
- o) Inspector 1: After the all clear interval, REMOVE Zone 2A and 2B kirk keys.
- p) Inspector 2: Break lockup using the device indicated in the table.
- q) Inspector 1 & 2: VERIFY the status of the devices as indicated in Table 22.
- r) Inspector 1 & 2: Perform the lockup of zones 1-4 with Inspector 2 remaining inside Zone 2B.
- s) Inspector 1: AT P1021.3-18 Insert Zones 1-4 Kirk Keys and TURN.
- t) Inspector 2: VERIFY:
 - a. Zone 2B All Clear horns are sounding _____
- u) Inspector 1: After the all clear interval, REMOVE Zone 2B, 3 and 4 kirk keys.

- v) Inspector 2: Break lockup using the device indicated in the table.
- w) Inspector 1 & 2: VERIFY the status of the devices as indicated in Table 23.
- x) Inspector 1 & 2: Perform the lockup of zones 1-4 with Inspector 2 remaining inside Zone 3.
- y) Inspector 1: AT P1021.3-18 Insert Zones 1-4 Kirk Keys and TURN.
- z) Inspector 2: VERIFY:
 - a. Zone 3 All Clear horns are sounding _____
- aa) Inspector 1: After the all clear interval, REMOVE Zone 3 and 4 kirk keys.
- bb) Inspector 2: Break lockup using the device indicated in the table.
- cc) Inspector 1 & 2: VERIFY the status of the devices as indicated in Table 24.
- dd) Inspector 1 & 2: Perform the lockup of zones 1-4 with Inspector 2 remaining inside Zone 4.
- ee) Inspector 1: AT P1021.3-18 Insert Zones 1-4 Kirk Keys and TURN.
- ff) Inspector 2: VERIFY:
 - a. Zone 4 All Clear horns are sounding _____
- gg) Inspector 1: After the all clear interval, REMOVE Zone 4 kirk key.
- hh) Inspector 2: Break lockup using the device indicated in the table.
- ii) Inspector 1 & 2: VERIFY the status of the devices as indicated in Table 25.
- jj) Inspector 1 & 2: Perform the lockup of zone 5, with Inspector 2 remaining inside Zone 5.
- kk) Inspector 1: At P1021.3-18 Insert Zone 5 Kirk Key and TURN.
- ll) Inspector 2: VERIFY:
 - a. Zone 5 All Clear horns are sounding _____
- mm) Inspector 1: After the all clear interval, REMOVE Zone 5 kirk key.
- nn) Inspector 2: Break lockup using the device indicated in the table.
- oo) Inspector 1 & 2: VERIFY the status of the devices as indicated in Table 26.
- pp) Inspector 1 or 2: In R1021.1-09, INSERT and TURN Gun Key #2 on.
- qq) Inspector 1 or 2: In R1032.2-01, INSERT and TURN RF Drive Key on.
- rr) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

5.7 Lockup Compromise Interlocks

NOTE: This section refers to all of the steps to the right of the “Area Compromise” banners of Tables 21 – 26.

NOTE: Inspector 2 must be inside the enclosure when locked to verify some components and is to have appropriate hearing protection.

NOTE: The entire lockup of zones 1-5 [8] must be performed before beginning these tests. After each test, the section must be locked again and the keys turned on in the control room before the next, allowing enough time for the all-clear horns to finish.

NOTE: To compromise an area, open the zone entrance for that area using the spare key for that door. Issuing these keys and ensuring their return to safe storage is the responsibility of the HSE department.

- a) Inspector 1 or 2: In R1021.1-09, TURN Gun Key #2 off and REMOVE Key.
- b) Inspector 1 or 2: In R1032.2-01, TURN RF Drive Key off and REMOVE Key.
- c) Inspector 2: Perform the lockups of the various zones [1-5] as shown in the Tables 21 – 26.
- d) Inspector 1: As each stage is completed as indicated by the checklist, VERIFY the status of the devices as indicated in Tables 21 - 26.
- e) Inspector 1 or 2: In R1021.1-09, INSERT and TURN Gun Key #2 on.
- f) Inspector 1 or 2: In R1032.2-01, INSERT and TURN RF Drive Key on.
- g) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

Zone 1 Normal and Compromise Tests																											
Stage		Before Lockup	STN #1	Exit Gate (G3)	STN #2	STN #3	STN #4	STN #5	Exit Door (D4)		D1		Before Test	G1	G2	G3	G4	G5	D1	D2	D3	D4	RD1	EOS 0002-01			
Device																											
PLC0102.5-01 Module 4 Ch 15	Normal Lockup	0	1	0	0	0	0	0	0	Break Lockup		Area Compromise															
PLC0102.5-01 Module 4 Ch 14		0	0	0	1	0	0	0	0																		
PLC0102.5-01 Module 4 Ch 17		0			0	1	0	0	0																		
PLC0102.5-01 Module 4 Ch 16		0				0	1	0	0																		
PLC0102.5-01 Module 4 Ch 18		0					0	1	0																		
Area Horns		0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Zone 1 Lockup/Secure Lights		0	0	0	0	0	0	0	0				1	0	1	0 ^{2B}	0	0	0 ^{2A}	0	0 ^{2A}	0	0	0	0	0 ^{2B}	0
P1021.3-18 Zone 1 Secure Light		0	0	0	0	0	0	0	0				1	0	1	0 ^{2B}	0	0	0 ^{2A}	0	0 ^{2A}	0	0	0	0	0 ^{2B}	0
P1021.3-18 EOS light		0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
P1021.3-18 Horn		0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
P1021.3-18 Major Fault Light		0	0	0	0	0	0	0	0				0	0	0	0	1	1	1	1	1	1	1	1	0	1	1
R1021.4-04 K6, K17-22, K25-28												1	0	0	0	0	0	0	0	0	0	0	0	0			
Mirror M0001-01																											
Mirror M0002-01																											

0=off, 1= on NOTE: PLC lights will only toggle when LUS is pressed.

2A = Also check Zone 2A lights

2B = Also check Zone 2B lights

Table 21: Zone 1 Normal and Compromise tests

Zone 2A Normal and Compromise tests (with Zone 2B locked up)																												
Stage		Before Lockup	Zone 2a STN #1	Zone 2a STN #2	Exit Gate (G8)	Zone 2a STN #3	Exit Gate (G6)	Zone 2a STN #4	Zone 2a STN #5	Zone 2a STN #6	Exit Gate (D5)	Zone 2a STN #7	Zone 2a Exit		G7		Before Test	G10	D8	G8	D5	G7	G6	G12	EOS 0006-02			
Device																												
PLC0102.5-01 Module 7 Ch 3	Normal Lockup	0	1	0	0	0	0	0	0	0	0	0	0	Break Lockup														
PLC0102.5-01 Module 7 Ch 4		0	0	1	0	0	0	0	0	0	0	0	0															
PLC0102.5-01 Module 7 Ch 9		0		0	0	1	0	0	0	0	0	0	0															
PLC0102.5-01 Module 7 Ch 0		0				0	0	1	0	0	0	0	0															
PLC0102.5-01 Module 7 Ch 1		0						0	1	0	0	0	0															
PLC0102.5-01 Module 7 Ch 2		0							0	1	0	0	0															
PLC0102.5-01 Module 7 Ch 5		0								0	0	1	0															
Area Horns		0	0	0	0	0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0	
Zone 2A Lockup/Secure Lights		0	0	0	0	0	0	0	0	0	0	0	0				1	0	0	0	0	0	0	0	0	0	0	
P1021.3-18 Zone 2A Secure light		0	0	0	0	0	0	0	0	0	0	0	0				1	0	0	0	0	0	0	0	0	0	0	0
P1021.3-18 EOS light	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1				
P1021.3-18 Horn	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1				
P1021.3-18 Major Fault Light	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1			
R1021.4-04 K6, K17-22, K25-28																		1	0	0	0	0	0	0	0	0	0	0
Mirror M0006-01																												

0=off, 1= on NOTE: PLC lights will only toggle when LUS is pressed.

2B = Also check Zone 2B lights

4 = Also check Zone 4 lights

Table 22: Zone 2A Normal and Compromise tests

Zone 2B Normal and Compromise tests (with Zone 2A locked up)																	
Device	Stage	Before Lockup	Zone 2b STN#1	Zone 2b STN#2	Zone 2b STN#3	Zone 2b STN#4	Zone 2b STN#5	Zone 2b Exit		G11		Before Test	G11	D9	EOS0003-01		
PLC0102.5-01 Module 8 CH 10	Normal Lockup	0	1	0	0	0	0	0	Break Lockup		Area Compromise						
PLC0102.5-01 Module 8 CH 11		0	0	1	0	0	0	0									
PLC0102.5-01 Module 8 CH 12		0		0	1	0	0	0									
PLC0102.5-01 Module 8 CH 13		0			0	1	0	0									
PLC0102.5-01 Module 8 CH 14		0				0	1	0									
Area Horns		0	0	0	0	0	0	0				0	0	0	0		
Zone 2B Lockup/Secure Lights		0	0	0	0	0	0	0				1	0	1	0 ⁴	0	0
P1021.3-18 Zone 2B Secure light		0	0	0	0	0	0	0				1	0	1	0 ⁴	0	0
P1021.3-18 EOS light		0	0	0	0	0	0	0				0	0	0	0	0	1
P1021.3-18 Horn		0	0	0	0	0	0	0				0	0	0	0	0	1
P1021.3-18 Major Fault Light	0	0	0	0	0	0	0	0	0	0	1	0	1				
R1021.4-04 K6, K17-22, K25-28												1	0	0	0		

0=off, 1= on NOTE: PLC lights will only toggle when LUS is pressed.

4 = Also check Zone 4 lights

Zone 3 Normal and Compromise tests														
Stage		Before Lockup	STN #1	Exit Door (D7)	STN #2	Exit Gate (G13)		G9		Before Test	D7	G13	G9	EOS 0012-02
Device														
PLC0102.5-02 Module 4 Ch 14	Normal Lockup	0	1	0	0	0	Break Lockup		Area Compromise					
PLC0102.5-02 Module 4 Ch 15		0	0	0	1	0								
Area Horns		0	0	0	0	0		0		0	0	0		
Zone 3 Lockup /Secure Lights		0	0	0	0	1		0		1	0	0 ⁴	0	
P1021.3-18 Zone 3 Secure light		0	0	0	0	1		0		1	0	0 ⁴	0	
P1021.3-18 EOS light		0	0	0	0	0		0		0	0	0	1	
P1021.3-18 Horn		0	0	0	0	0		0		0	0	0	1	
P1021.3-18 Major Fault light		0	0	0	0	0		0		0	1	0	1	1
R1021.4-04 K6, K17-22, K25-28										1	0	0	0	0

0=off, 1= on NOTE: PLC lights will only toggle when LUS is pressed.

4 = Also check Zone 4 lights

Table 24: Zone 3 Normal and Compromise tests

Zone 4 Normal and Compromise tests																							
Device	Stage	Before Lockup	STN #1	STN #2	Exit Gate (G6)	STN #3	STN #4	STN #5	Exit Door (D6)	STN #6	Exit Door (D10)		G16		Before Test	G6	G16	D6	D10	EOS 0004-01			
PLC0102.5-02 Module 6 Ch 13	Normal Lockup	0	1	0	0	0	0	0	0	0	0	Break Lockup		Area Compromise									
PLC0102.5-02 Module 6 Ch 14		0	0	1	0	0	0	0	0	0	0												
PLC0102.5-02 Module 6 Ch 12		0		0	0	1	0	0	0	0	0												
PLC0102.5-02 Module 6 Ch 10		0				0	1	0	0	0	0												
PLC0102.5-02 Module 6 Ch 11		0					0	1	0	0	0												
PLC0102.5-02 Module 6 Ch 15		0						0	0	1	0												
Area Horns		0	0	0	0	0	0	0	0	0	0				0	0	0	0	0	0	0		
Zone 4 Lockup/Secure Lights		0	0	0	0	0	0	0	0	0	0				1	0	1	0	0	0 ⁵	0	0	0
P1021.3-18 Zone 4 Secure light		0	0	0	0	0	0	0	0	0	0				1	0	1	0	0	0 ⁵	0	0	0
P1021.3-18 EOS light		0	0	0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	1
P1021.3-18 Horn	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1				
P1021.3-18 Major Fault light	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1			
R1021.4-04 K6, K17-22, K25-28															1	0	0	0	0	0	0		
R1021.4-04 K7, K15-16, K29															1	1	0	1	1	1	1		

0=off, 1= on NOTE: PLC lights will only toggle when LUS is pressed.

5 = Also check Zone 5 lights

Table 25: Zone 4 Normal and Compromise tests

Zone 5 Normal and Compromise Tests																			
Device	Stage	Before Lockup	STN #1	STN #2	Exit Gate (G18)	STN #3	STN #4	STN #5	STN #6	Exit Door (G21)		G16		Before Test	G18	G20	G19	G21	EOS 0108-01
PLC0102.5-02 Module 8 Ch 19	Normal Lockup	0	1	0	0	0	0	0	0	0	Break Lockup		Area Compromise						
PLC0102.5-02 Module 8 Ch 17		0	0	1	0	0	0	0	0	0									
PLC0102.5-02 Module 8 Ch 14		0		0	0	1	0	0	0	0									
PLC0102.5-02 Module 8 Ch 16		0				0	1	0	0	0									
PLC0102.5-02 Module 8 Ch 18		0					0	1	0	0									
PLC0102.5-02 Module 8 Ch 15		0						0	1	0									
Area Horns		0	0	0	0	0	0	0	0	0		0		0	0	0	0	0	0
Zone 5 Lockup/Secure Lights		0	0	0	0	0	0	0	0	0		1		0	1	0	0	0	0
P1021.3-18 Zone 5 Secure light		0	0	0	0	0	0	0	0	0		1		0	1	0	0	0	0
P1021.3-18 EOS light		0	0	0	0	0	0	0	0	0		0		0	0	0	0	0	1
P1021.3-18 Horn		0	0	0	0	0	0	0	0	0		0		0	0	0	0	0	1
P1021.3-18 Major Fault light		0	0	0	0	0	0	0	0	0		0		0	0	1	1	1	0
R1021.4-04 K6-7, K15-22, K25-29													1	0	0	0	0	0	
Mirror M1301-01																			
Mirror M1304-01																			

0=off, 1= on NOTE: PLC lights will only toggle when LUS is pressed.

Table 26: Zone 5 Normal and Compromise tests

5.8 In Progress Interruptions

NOTE: For this section, refer to Table 27 to Table 32.

- a) Inspector 1 or 2: ENSURE accelerator radiation sources are disabled and cannot be turned on (for example, obtain electron gun key and RF source key).
- b) Inspector 2: For each column in the tables, ENSURE the area is ready for lock up search. Prior to the test, the first lock up station in the area should be flashing. Close doors, reset emergency off switches, and lock up required areas as necessary.
- c) Inspector 2: INTERRUPT the lock up using the method/device noted on the table. Refer to the notes below for additional information.
- d) Inspector 1: VERIFY the devices and place a checkmark on the table.
- e) Inspector 1 and 2: REPEAT the above steps for all of the columns on the tables.
- f) Inspectors 1 & 2: Print your names, date the checklist and initial where indicated.

Emergency Off: Activate any emergency off switch within the zone, and verify the lock up search cannot start.

Gate (G)/Door (D)/Radiation Door (RD)/Kirk Key (KKS): Open the door/gate/switch and verify the lock up search cannot start.

Area Timer: A lock up search will reset, if the area timer runs out before the lock up search is complete. Activate the first lockup station the area, and use a stopwatch to verify the time.

Exit Timer: For areas with more than one lock up station, the exit time is the allowable time between pressing the last lock up station and leaving the area. Perform the lock up search, up to and including activating the last lock up station, and use a stopwatch to verify the time.

LUS Out of Sequence: Lock up stations must be activated in the correct order. Attempt a lock up search of the area by activating each lock up station, but omitting the first lock up station. Verify the lock up search is not allowed to start.

Required Area: Most zones are comprised of multiple areas. Required areas must be searched prior to searching dependent areas. Verify the dependent area lock up search cannot start, by breaking lock up in the required area.

Inspector:_____	Initials:_____
(Please print)	Date (dd-mm-yyyy):_____
Inspector:_____	Initials:_____
(Please print)	Date (dd-mm-yyyy):_____

Zone 1 In-Progress Interruptions

Device	Stage	Emergency Off (1)	G1 (1.1)	G2 (1.1)	RD1 (1.1)	G3 (1.1)	Area Timer (1.1, 15 s)	Required Area (1.1)	G5 (1.2)	D1 (1.2)	G4 (1.2)	D2 (1.2)	Area Timer (1.2, 60 s)	Exit Timer (1.2, 20 s)	LUS Out of Seq. (1.2)	Required Area (1.2)	D3 (1.3)	D4 (1.3)	Area Timer (1.3, 30 s)
LUS #1 Lamp		0	0	0	0	0	*												
LUS #2 Lamp								0	0	0	0	0	*	*	*				
LUS #3 Lamp													0	0	0				
LUS #4 Lamp													0	0	0				
LUS #5 Lamp																0	0	0	*

Table 27: Zone 1 In-Progress Interruptions

* = Flashing

Zone 2A In-Progress Interruptions

Stage																							
Device	Emergency Off (2A)	G10 (2A.1)	D8 (2A.1)	KKSW00006-01 (2A.1)	G8 (2A.1)	Area Timer (2A.1, 20 s)	Exit Timer (2A.1, 15 s)	LUS Out of Seq. (2A.1)	G6 (2A.2)	Area Timer (2A.2, 10 s)	Required Area (2A.1)	Required Area (2A.2)	D1 (2A.3)	G4 (2A.3)	G7 (2A.3)	D5 (2A.3)	Area Timer (2A.3, 40 s)	Exit Timer (2A.3, 10 s)	LUS Out of Seq. (2A.3)	Required Area (2A.3)	G12 (2A.4)	Area Timer (2A.4, 40 s)	
LUS #1 Lamp	0	0	0	0	0	*	*	*															
LUS #2 Lamp						0	0	0															
LUS #3 Lamp	0								0	*													
LUS #4 Lamp											0	0	0	0	0	0	*	*	*				
LUS #5 Lamp																	0	0	0				
LUS #6 Lamp																	0	0	0				
LUS #7 Lamp																				0	0	*	

Table 28: Zone 2A In-Progress Interruptions

* = Flashing

Zone 2B In-Progress Interruptions

Device	Stage	Emergency Off (2B)	G7 (2B)	G11 (2B)	RD1 (2B)	G1 (2B)	D9 (2B)	Area Timer (2B, 65 s)	Exit Timer (2B, 30 s)	LUS Out of Seq. (2B)
LUS #1 Lamp		0	0	0	0	0	0	*	*	*
LUS #2 Lamp								0	0	0
LUS #3 Lamp								0	0	0
LUS #4 Lamp								0	0	0
LUS #5 Lamp								0	0	0

Table 29: Zone 2B In-Progress Interruptions

* = Flashing

Zone 3 In-Progress Interruptions

Device	Stage	Emergency Off (3)	D7 (3.1)	Area Timer (3.1, 20 s)	Required Area (3.1)	G9 (3.2)	G13 (3.2)	Area Timer (3.2, 40 s)
LUS #1 Lamp	0	0	*					
LUS #2 Lamp				0	0	0	*	

Table 30: Zone 3 In-Progress Interruptions

* = Flashing

Zone 4 In-Progress Interruptions

Device	Stage	Emergency Off (4)	G6 (4.1)	Area Timer (4.1, 40 s)	Exit Timer (4.1, 15 s)	LUS Out of Seq. (4.1)	Required Area (4.1)	G9 (4.2)	G10 (4.2)	G11 (4.2)	D6 (4.2)	G16 (4.2)	Area Timer (4.2, 60 s)	Exit Timer (4.2, 20 s)	LUS Out of Seq. (4.2)	Required Area (4.2)	D10 (4.3)	Area Timer (4.3, 15 s)
LUS #1 Lamp		0	0	*	*	*												
LUS #2 Lamp				0	0	0												
LUS #3 Lamp							0	0	0	0	0	0	*	*	*			
LUS #4 Lamp													0	0	0			
LUS #5 Lamp													0	0	0			
LUS #6 Lamp																0	0	*

Table 31: Zone 4 In-Progress Interruptions

* = Flashing

Zone 5 In-Progress Interruptions

Device	Stage	Emergency Off (5)	G16 (5.1)	G18 (5.1)	Area Timer (5.1, 35 s)	Exit Timer (5.1, 10 s)	LUS Out of Seq. (5.1)	Required Area (5.1)	G19 (5.2)	G20 (5.2)	G21 (5.2)	Area Timer (5.2, 40 s)	Exit Timer (5.2, 10 s)	LUS Out of Seq. (5.2)
LUS #1 Lamp		0	0	0	*	*	*							
LUS #2 Lamp					0	0	0							
LUS #3 Lamp								0	0	0	0	*	*	*
LUS #4 Lamp												0	0	0
LUS #5 Lamp												0	0	0
LUS #6 Lamp												0	0	0

Table 32: Zone 5 In-Progress Interruptions

* = Flashing

5.9 Radiation Sources Verification

Note: Verification of the status of the radiation sources will require deferring to operational staff. It is up to the inspectors to challenge the assertions until they are satisfied with the state of the components.

- a) Inspector 1: Complete the Personnel Safety System Jumper Request form (11.11.52.16), open a service request, and obtain controlled work permit for an electrical technician to install the jumper, as noted in step k) below.
- b) Inspectors 2 and 3: ENSURE that all of zones 1-9 are locked [8] [36] [37] up to and including turning all gate keys in the control room and allowing the respective All Clear intervals to time out.
- c) Inspector 1: OBTAIN the assistance of someone from AOD as indicated in 3.2, hereinafter referred to as "AOD Assistant".
- d) Inspector 1: OBTAIN the assistance of someone from Engineering as indicated in 3.2, hereinafter referred to as "Engineering Assistant".
- e) Inspector 1: ENSURE all Linac Ready Chains are enabled.
- f) Inspector 1: ENSURE LTB beam stop and BTS collimator and scraper are open.

SECTION "A" (BYPASSES OFF)

- g) Inspectors 2 and 3: At the panel indicated in Table 33, PRESS SWx.

Note: The LINAC ACIS panels are located in the power supply room (0102.3).

Note: P1609-06 is the QMSC ACIS panel. Ensure the hutch door is open.

- h) Inspector 1 and AOD Assistant: VERIFY the status of the SOURCES for the SW pressed as per Table 33.
- i) Inspector 2: Lock up zone 5 if required (PLC test only).
- j) Inspector 1 and AOD Assistant: Repeat steps g) to i) for remaining columns in section "A".

SECTION "B" (BYPASSES ON)

Note: In this section, bypasses are required to ensure the ACIS has interlocked the radiation sources. Otherwise, the machine protection system would also interlock the radiation sources.

- k) Electrical Technician: Print name in the 'JUMPERED BY' section of the Personnel Safety System Jumper Request Form, and place form in 'Active' section of Personnel Safety System Jumper Log binder in control room. INSTALL jumper between TB1-6 and TB1-7 at P1021.4-16 in R1021.4-03. Refer to drawings 0055537 and 0130390.
- l) Engineering Assistant: Log into the LINAC machine protection PLC (PLC0009-02) and FORCE the following variables:

<u>Address</u>	<u>Forced Value</u>	<u>Variable</u>
100047	1	ACIS_BR1_SR1_Permissive_SUM
100111	1	ACIS_Linac_Permissive_PLC
100112	1	ACIS_Linac_Permissive_HW
100046	1	Linac_BeamOn_Button

- m) Inspector 1 and AOD Assistant: At P1021.1-03, place the Gun HV in manual control.
- n) Inspector 1 and AOD Assistant: Set Gun Voltage to 25 kV.
- o) Inspector 1 and AOD Assistant: VERIFY the status of the SOURCES in the 'BEFORE TEST' column as per Table 33.
- p) Inspectors 2 and 3: At Panel indicated in Table 33, PRESS SWx.
- q) Inspector 1 or Electrical Technician and AOD Assistant: VERIFY the status of the SOURCES for the SW pressed as per Table 33.
- r) Inspector 2: Lock up the last zone tested, if required. (N/A for Zone 5 HW and QMSC test switches)
- s) Inspector 1 and AOD Assistant: RESET radiation sources as required.
- t) Inspector 1 and AOD Assistant: Repeat steps n) through s) for remaining columns in section "B".
- u) Inspector 1: After successful completion of Section "B", print name in the "TESTED BY" section of the Personnel Safety System Jumper Request form (11.11.52.16).

SECTION "C" (BYPASSES OFF)

- v) Engineering Assistant: REMOVE the forces on the LINAC machine protection PLC (PLC0009-02) variables:

Address	Variable
100047	ACIS_BR1_SR1_Permissive_SUM
100111	ACIS_Linac_Permissive_PLC
100112	ACIS_Linac_Permissive_HW
100046	Linac_BeamOn_Button

- w) Electrical Technician: REMOVE jumper between TB1-6 and TB1-7 at P1021.4-16 in R1021.4-03. Print name in the 'JUMPER REMOVED BY' section of the Personnel Safety System Jumper Request Form. Leave the form in 'Active' section of Personnel Safety System Jumper Log binder in control room.
 - x) Inspector 1 and AOD Assistant: At P1021.1-03, place the Gun HV in manual control.
 - y) Inspector 1 and AOD Assistant: Set Gun Voltage to 25 kV.
 - z) Inspectors 2 and 3: At Panel indicated in Table 33, PRESS SWx.
 - aa) Inspector 1 or Electrical Technician and AOD Assistant: VERIFY the status of the SOURCES for the SW pressed as per Table 33.
 - bb) Inspector 2: Lock up zone 5 if required (PLC test only).
 - cc) Inspector 1 and AOD Assistant: RESET radiation sources as required.
 - dd) Inspector 1 and AOD Assistant: Repeat steps y) through cc) for remaining columns in section "C".

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NOTE 1: Observe that the LINAC Master Display Screen indicator “RF Enable” is RED.

NOTE 2: From the LINAC machine protection PLC (PLC0009-02), observe the LINAC “personnel protection” latch status from the ‘Linac_PersProt’ variable at address 100093.

NOTE 3: Looking at the gun HV digital readout, observe that the gun HV turns off. “0” indicates voltage is zero (shows 000) immediately (crowbars) and does not ramp down slowly. “R” indicates a rundown, which will ramp the voltage from 25 kV to zero in approximately 4 seconds.

NOTE 4: Observe that the gun HV ON (RED) lamp turns off. In the case of a rundown (“R”), the lamp will turn off after the run down is complete.

NOTE 5: Observe that the gun BEAM ON (RED) lamp turns off.

NOTE 6: Observe that the RF ON light at the RF Source (R1032.2-01) turns OFF.

NOTE 7: Use a ladder to observe the LINAC RF Trigger board, located on top of the RF source in room 1032. Observe that the interlocks ok LED “DS1” turns from green to off.

NOTE 8: Observe that the gun coax switch is interlocked. ENSURE the voltage from TB0-2 to TB0-3 in R1021.2-09 falls from 24 V to 0 V.

NOTE 9: Observe that the Booster Ring RF HVPS screen indicates CLS External Interlock (safety) is RED.

NOTE: PRESS AND HOLD ALL TEST SWITCHES

**** Zone 5 will need to be locked again after pressing test switch.**

R = RUNDOWN

	A (bypasses off)				B (bypasses on)										C (bypasses off)			
	Zone 5 (PLC – P0102.5-03 – SW40**)	Zone 5 (HW – P0102.5-03 – SW39)	Zones 6-9+BL (HW – P1609-06 – SW9)	Zones 6-9+BL (PLC – P1609-06 – SW15)	BEFORE TEST	Zone 1 (P0102.5-01 – SW1)	Zone 2A (P0102.5-02 – SW18)	Zone 2B (P0102.5-02 – SW53)	Zone 3 (P0102.5-03 – SW4)	Zone 4 (P0102.5-03 – SW25)	Zone 5 (PLC – P0102.5-03 – SW40**)	Zone 5 (HW – P0102.5-03 – SW39)	Zones 6-9+BL (HW – P1609-06 – SW9)	Zones 6-9+BL (PLC – P1609-06 – SW15)	Zone 5 (PLC – P0102.5-03 – SW40**)	Zone 5 (HW – P0102.5-03 – SW39)	Zones 6-9+BL (HW – P1609-06 – SW9)	Zones 6-9+BL (PLC – P1609-06 – SW15)
1. RF Enable (on LINAC Master Display Screen)	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0
2. LINAC “Personnel Protection”	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	0
3. Gun HV Digital Readout					1	0	0	0	0	0	0	0	R	R	0	0	R	R
4. Linac Gun HV Off/On					1	0	0	0	0	0	0	0	R	R	0	0	R	R
5. Linac Beam Off/On					1	0	0	0	0	0	0	0	0	0	0	0	0	0
6. Linac RF (at RF Source)					1	0	0	0	0	0	0	0	0	0	0	0	0	0
7. LINAC RF Drive Trigger (at RF Source)					1	0	0	0	0	0	0	0	0	0	0	0	0	0
8. Gun Coax Switch					1	0	0	0	0	0	0	0	0	0	0	0	0	0
9. BR RF Enable (on BR1 Master Display Screen)					1	1	1	1	1	1	0	0	0	0	0	0	0	0

Table 33: Radiation Sources Verification

- ee) Inspector 1: After successful completion of Section "C", print name in the "PSS FUNCTION TEST BY" section of the Personnel Safety System Jumper Request form (11.11.52.16) and place sheet in 'Complete' section of Personnel Safety System Jumper Log binder in control room.
- ff) All Inspectors and Assistants: Print your names, date the checklist and initial where indicated.

AOD Assistant: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Engineering Assistant: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Inspector: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____
Electrical Technician: _____	Initials: _____
(Please print)	Date (dd-mm-yyyy): _____

6.0 FORMS

6.1 Non-Conformance Summary

Date(s) of Validation: _____

CLS Problem ID

Cc: HSE Manager
CLS Files

Appendix A - Removal and Installation of Mini-Relays

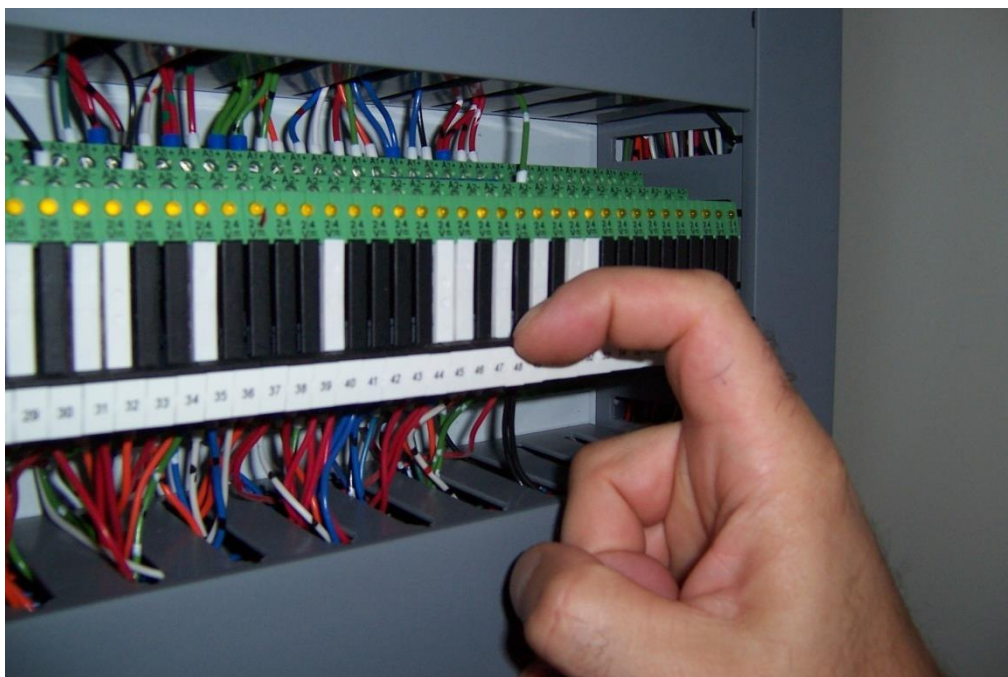


Figure 1: Press down as shown to unlock relay...

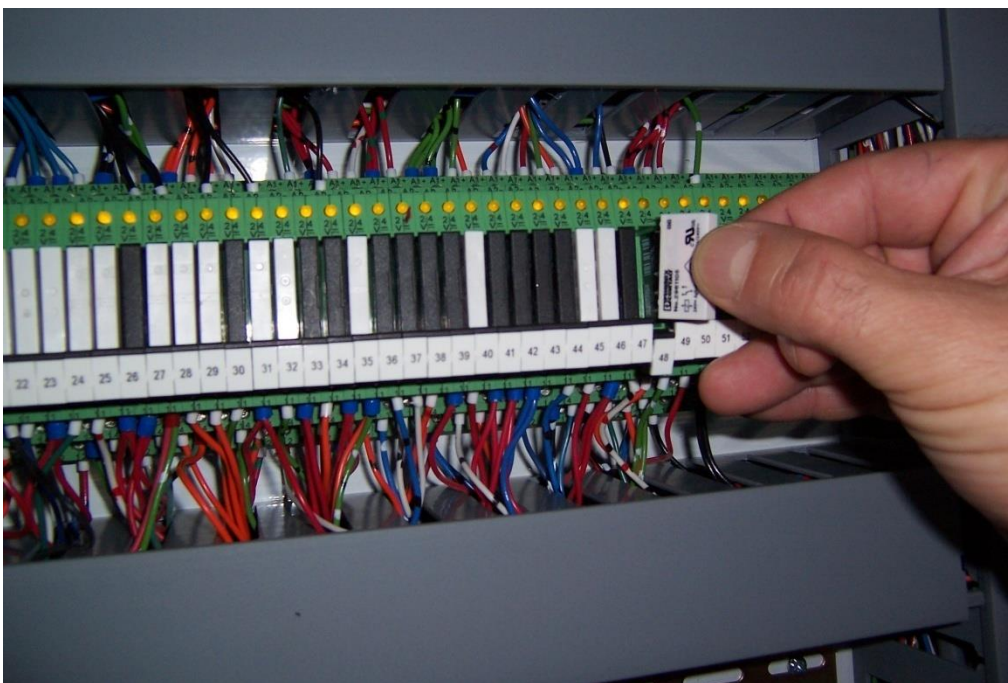


Figure 2:...and remove relay.

Installation is the reverse of removal. Ensure pins line up with the holes before inserting in to base.

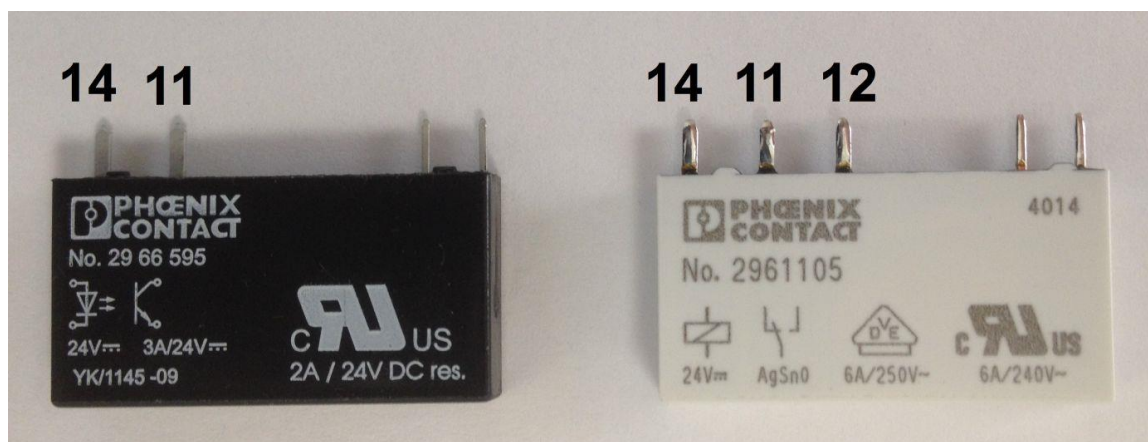


Figure 3: Relay Pinouts

Appendix B - Resetting PLC

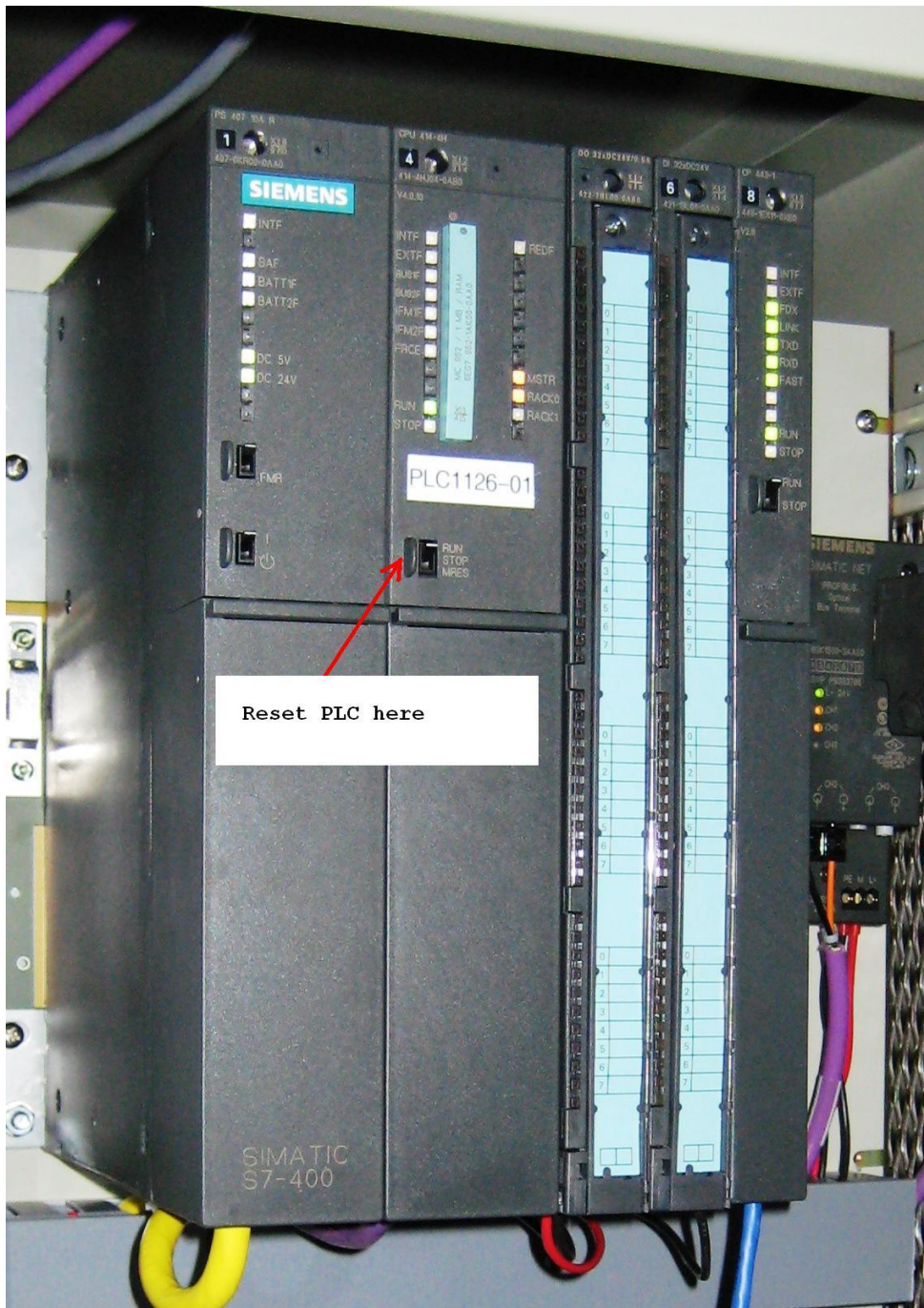


Figure 4: Resetting PLC

REVISION HISTORY

Revision	Date	Description	Author
A	2009-Apr-21	Initial Draft	C. Finlay
0	2009-May-26	Issued for use	C. Finlay
0A	2009-Oct-06	Updated with as built drawing changes. Added co2 system relays.	C. Finlay
0B	2009-Oct-08	Updated with Reviewer comments.	C. Finlay
1	2009-Oct-08	Issued for Use	C. Finlay
2	2009-Oct-09	Issued for use	C. Finlay
2A	2010-May-05	MKS 6220, 6221, 6823	C. Finlay
3	2010-Aug-31	Issued for use	C. Finlay
3A	2010-Sep-08	MKS 6223, 6736, ECO638	C. Finlay
4	2010-Sep-21	Issued for use	C. Finlay
4A	2010-Nov-02	MKS 7110, 7112, 7124, 7125, 7126, 7164	C. Finlay
4B	2010-Nov-23	Incorporated changes from review of 4a	C. Finlay
5	2010-Dec-15	Issued for use	C. Finlay
5A	2011-Jan-20	Changes to Zones 2/3 for EA2	C. Finlay
6	2011-Feb-01	Issued for use	C. Finlay
		Incorporated changes found during V+V (MKS 7887)	
6A	2011-Sep-15	Added 2 nd reviewer	B. Schneider
7	2011-Sep-28	Issued for use	B. Schneider
7A	2012-Nov-06	MKS8249, MKS8504	B. Schneider
8	2012-Nov-07	Updated reviewers, Issued for use	B. Schneider
8A	2013-Sep-10	MKS8813, MKS8811	B. Schneider
9	2013-Oct-03	Issued for use	B. Schneider
9A	2014-Sep-16	MKS8811, MKS9147	B. Schneider
10	2014-Oct-06	MKS9158. Issued for use.	B. Schneider
10A	2015-Sep-23	MKS9443	B. Schneider
11	2015-Oct-15	Issued for use	B. Schneider
11A	2016-May-05	MKS9542, MKS9587	B. Schneider

12	2016-Jun-01	Incorporated comments, issued for use	B. Schneider
12A	2017-Jan-10	MKS9742	B. Schneider
12B	2017-Jan-18	Incorporated comments	B. Schneider
12C	2017-Feb-08	MKS9772	B. Schneider
13	2017-Feb-14	Issued for use	B. Schneider
13A	2017-Apr-04	MKS9808	B. Schneider
14	2017-Apr-05	Incorporated comments, issued for use	B. Schneider
14A	2017-Oct-18	MKS9936, MKS9941	B. Schneider
15	2017-Oct-31	Issued for use	B. Schneider
15A	2018-Mar-15	MKS10034, MKS10026	B. Schneider
16	2018-Mar-21	Issued for use	B. Schneider
16A	2018-Oct-02	MKS10141	B. Schneider
17	2018-Oct-05	Incorporated comments, issued for use	B. Schneider
17A	2018-Nov-02	MKS10175	B. Schneider
18	2018-Nov-26	Issued for use	B. Schneider